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Deep multi-view least squares support vector machine with consistency and complementarity principle based on cross-output knowledge transfer

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ABSTRACT

This article presents a Deep Multi-view Least Squares Support Vector Machine with Consistency and Complementarity Principles based on Cross-Output Knowledge Transfer (MDCTM), which has four distinctive features: 1) It integrates the idea of deep stacking architecture, which is the first attempt to use transfer learning to form deep architectures in multi-view learning. It can enhance the ability to handle complex problems. Starting from the second layer, it incorporates extra input attributes that consider the predictions made by all preceding layers, effectively revealing the manifold structure of the original data. 2) Each layer follows the consistency and complementarity principles, which can fully excavate the information in multi-view data. In each layer, the model is solved by an alternating optimization strategy. 3) Cross-output knowledge transfer leverages predictions from earlier layers to improve the learning of subsequent ones, which can improve the classification performance of the model. Additionally, the extent of cross-output knowledge transfer between sequential layers can be assessed autonomously and effectively by utilizing a fast leave-one-out cross-validation method. 4) The model allows random assignment of model parameters in each layer, such as weights and kernel widths, boosting learning speed. Numerical experiments demonstrate the model's effectiveness and efficiency.

1. Introduction

With the progress of computer technology, vast and intricate data have appeared in numerous fields. Multi-view data signifies feature sets obtained from multiple channels or layers for a single object (Li et al., 2018). Effectively extracting valuable information from multi-view data has become a key research focus in the fields of machine learning and data analytics. On this basis, multi-view learning (MVL) has emerged. In many fields, MVL has demonstrated its unique advantages (Zhao et al., 2022) (Xu et al., 2024) (Pandey et al., 2024).

Multi-view data comprises different views that are inherently related yet distinct. As a result, MVL generally adheres to two key principles: consistency and complementarity (2C). The consistency principle focuses on maximizing the shared information across various views. Conversely, the complementarity principle leverages the unique information from each view to enhance one another, thereby improving the method's performance and effectiveness. Guided by these principles, many excellent single-view models have also been expanded to the multiview domain. For instance, Sun and Chao (2013) proposed incorporating view-marginal consensus into the maximum entropy discrimination (MED) framework. Multiview classification with cohesion and diversity (MvCD) (Tao et al., 2020) builds upon discriminative least-squares regression (DLSR) (Xiang et al., 2012) by adapting it to the MVL paradigm, where its performance is improved by strengthening between-class cohesion and encouraging diversity across views.

As a mature single-view classification method, support vector machine (SVM) (Cortes & Vapnik, 1995) achieves classification by exploring the optimal hyperplane with the largest margin between two types of examples, showing outstanding performance. Currently, SVM has been integrated into MVL framework, which we collectively refer to SVM-based MVL. Farquhar et al. (2005) proposed SVM-2K by combining kernel canonical correlation analysis (KCCA) with SVM firstly, which incorporates a similarity constraint on one-dimensional projections to align the information from two different views and promote consistency. Because SVM-2K needs to solve a large-scale quadratic programming problem (QPP), the computational complexity is relatively high. Xie and Sun (2015) proposed multi-view twin SVM (MvTSVM). It solves two smaller-sized QPPs in two different feature spaces, which decreases the computational complexity. However, both SVM-2K and MvTSVM failed to uncover complementary information between multi-view data. To improve this drawback, MvSVM-2C (Xie & Sun, 2020) uses view weights to integrate information from different views. Multi-view privileged

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SVM (PSVM-2V) (Tang et al., 2017) implements the complementarity including the introduction of privileged information. Different from these thoughts, Tang et al. (2019) proposed coupling privileged kernel method for MVL (MCPK), its consistency is captured by the coupling term between two views, the implementation of complementarity is the same as PSVM-2V. Multi-view margin distribution model (MVLDM) (Hu et al., 2024) focuses on margin distribution, thereby fulfilling 2C principles. Multiview structural large margin classifier (MvSLMC) (Zhao & Xu, 2024) attempts to realize consistency by minimizing the compactness of clusters in all views, complementarity is embodied by view weights and structural information. Similarly, the multi-view SVM classifier via L0/1 soft-margin loss and structural information (Mv$SL_{0/1}$-SVM) (Chen et al., 2025), as well as the multi-view structural TSVM with the 2C principles (MvSTSVM) (Liu et al., 2025), both integrate structural information to further improve classification performance. However, the above models all need to solve large-scale QPPs, which is time-consuming, especially in the case of large samples. In contrast, multi-view LS-SVM (MV-LSSVM)(Houthuys et al., 2018) transforms the classification task into a closed-form solution for solving a system of linear equations, shortening the running time. MV-LSSVM only uses the coupling term to excavate the complementarity information in multi-veiw data. Consensus and complementarity-based MV-LSSVM (Tang et al., 2022) is proposed to satisfy both two principles which is implemented by minimizing the least squares loss of the inconsistency of predictions between different views and adaptively adjusting the weight of each views.

In summary, numerous advanced SVM models have been applied to the field of MVL using various approaches, some of which have demonstrated impressive performance. However, these models may still be constrained in certain aspects. Firstly, the generalization ability of the SVM-based MVL are greatly affected by the model hyperparameters, such as regularization parameters and kernel parameters. The traditional method to determine these hyperparameters is k-fold cross-validation. Leave-one-out cross-validation is usually used as the parameter selection strategy to ensure unbiased estimation and obtain optimal generalization ability. In reality, leave-one-out cross-validation is often highly time-intensive and requires substantial computational power. Therefore, it is necessary to propose a method to quickly select model parameters. Secondly, these models are shallow structures so have limited ability to learn complex functions for complex data. Trying to incorporate deep architectures (Yu & Deng, 2011) into learning frameworks may improve their classification performance. Finally, the most of these models cannot utilize previously learned knowledge for learning, which hinders the improvement of generalization ability. Transfer learning (Pan & Yang, 2010) just happens to fill this gap.

In order to integrate the advantages of deep stacking architecture and transfer learning, improve the generalization ability of SVM-based MVL in complex problems, and reduce the parameter search time, we propose MDCTM based on MV-LSSVM. We do experiments on 5 UCI datasets, 9 Handwritten datasets, and 45 AWA datasets, verifying the excellent performance of the MDCTM. The key contributions of this paper are outlined as follows.

1) We employ a deep stacking architecture to organize multiple modules in a layered structure, which is the first attempt to use transfer learning to form deep architectures in MVL. Starting from the second layer, each module utilizes the predicted outputs of all preceding modules, effectively revealing the manifold structure of the original data.

2) Each module follows the 2C principles. Minimizing the least squares loss of the training error and the inconsistency of predictions between different views satisfies the consistency principle, and the weight of each view is adjusted to satisfy the complementarity principle. The entire algorithm can be solved by an alternating optimization strategy. Therefore, each module can fully explore the consistency and complementary information of multi-view data.

3) Inspired by transfer learning, we train a weight parameter that quantifies the influence of prediction values from all preceding lower-level modules on the current module layer. Additionally, by employing a fast leave-one-out cross-validation method to estimate this parameter, the model can autonomously and efficiently determine the magnitude of cross-output knowledge transfer between contiguous modules. This adaptive mechanism significantly enhances the model's flexibility and generalization capabilities.

4) MDCTM allows random assignment of model parameters in each module, such as weights and kernel widths, significantly streamlining the learning process.

The subsequent sections of this paper are organized in the following manner. In Section 2, we briefly review deep stacking architectures, transfer learning, multi-view learning, and the MV-LSSVM model. In Section 3, we propose the MDCTM and provide theoretical derivations and algorithms. Section 4 provides theoretical analysis. Section 5 lists the results of numerical experiments. Finally, we draw conclusions and future work in Section 6.

2. Related works

In this section, we review the foundational research on deep architectures, transfer learning, multi-view learning, and the MV-LSSVM model.

2.1. Deep architectures

The deep learning architecture has the advantages of automatic feature learning, efficient processing of complex tasks, good scalability, and strong generalization capabilities. It is also significantly advantageous in processing large-scale data and complex nonlinear problems. Feedforward neural networks (Hornik et al., 1989), convolutional neural networks (Jadeja et al., 2023), autoencoders (Li et al., 2023), long short-term memory (Van Houdt et al., 2020), and generative adversarial networks (Gui et al., 2023) are all variants of deep architectures. They currently have made significant contributions to the development of the modern transportation industry (Salihoglu et al., 2024), image synthesis (Wang et al., 2024) (Smith et al., 2020), natural language processing (Shen et al., 2020) (Brown et al., 2021), biomedical science (Johnson et al., 2021) and multi-view clustering (MVC) (Du et al., 2023).

The deep architecture used in this study is derived from the deep stacking network (DSN) proposed by Yu and Deng (2011). DSN and stacked generalization (Wolpert, 1992) both have the core idea of enhancing the expressiveness or predictive performance of the model by stacking multiple layers or models. The difference is that DSN ensures that each layer learns new and more abstract features from the previous layer's output through layer-by-layer abstraction and training.

2.2. Transfer learning

Traditional machine learning algorithms usually require retraining the model for each new task, making it difficult to adapt quickly. Therefore, we introduce transfer learning, the core idea that the knowledge obtained in the source task (such as features and model weights) can boost the learning effectiveness when the target task involves limited data or presents significant training challenges.

In transfer learning, the definitions of domain and task are introduced. The domain $\mathbb{D}$ consists of two parts: feature space $\mathcal{X}$ and marginal probability distribution $\mathbb{P}^X$, where each input sample is $\boldsymbol{x} \in \mathcal{X}$. In general, two different domains have different feature spaces or marginal probability distributions. In a specific domain $\mathbb{D} = \{\mathcal{X}, \mathbb{P}^X\}$, the task $\mathbb{T} = \{\mathcal{Y}, f(\cdot)\}$ consists of label space $\mathcal{Y}$ and function $f(\cdot)$. The function $f(\cdot)$ is a prediction function that can predict labels for unknown samples $\{\boldsymbol{x}^*\}$. Specifically, we define $\mathcal{D}_s = \left\{\left(\boldsymbol{x}_{s_i}, y_{s_i}\right)\right\}_{i=1}^{n_s}$ as the labeled data in the source domain, where $\boldsymbol{x}_{s_i} \in \mathcal{X}$ and $y_{s_i} \in \mathcal{Y}$ are the data samples and the corresponding category labels, respectively. Similarly, $\mathcal{D}_t = \left\{\left(\boldsymbol{x}_{t_i}, y_{t_i}\right)\right\}_{i=1}^{n_t}$ is the labeled data in the target domain, where $\boldsymbol{x}_{t_i} \in \mathcal{X}_t$ and $y_{t_i} \in \mathcal{Y}_t$ are the input and output in the target domain, respectively.

According to the above definition, given the source domain $\mathbb{D}_s$ and learning task $\mathbb{T}_s$, the target domain $\mathbb{D}_t$ and learning task $\mathbb{T}_t$. The purpose of transfer learning is to obtain knowledge in the source domain $\mathbb{D}_s$ and learning task $\mathbb{T}_s$ to help improve the learning of the prediction function $f_t(\cdot)$ in the target domain, where $\mathbb{D}_s \neq \mathbb{D}_t$ or $\mathbb{T}_s \neq \mathbb{T}_t$.

Building on the definition provided above, existing transfer learning methods can be categorized in various ways. Specifically, depending on whether the feature space and/or label space are isomorphic, transfer learning is divided into two types: isomorphic transfer learning and heterogeneous transfer learning (Torrey & Shavlik, 2009). Weiss et al. (2016) provided a comprehensive summary of isomorphic and heterogeneous transfer learning methods, which have since found practical applications in various domains (Chui et al., 2023). Taking into account the availability of labeled data in the target domain, existing transfer learning methods can be classified into three categories: supervised transfer learning, semi-supervised transfer learning, and unsupervised transfer learning (Torrey & Shavlik, 2009). From the perspective of "how to transfer", it is divided into instance-based transfer, feature-based transfer, parameter-based transfer, and relationship-based transfer (Weiss et al., 2016). Among them, instance-based transfer is mainly based on instance weighting strategy, feature-based transfer is based on the shared subspace of features in the source domain and the target domain, parameter-based transfer uses model/parameter to transfer knowledge, and relationship-based transfer suggests that certain connections between instances are analogous between tasks. Some surveys have studied the progress of transfer learning in specific contexts, such as reinforcement learning (Zhu et al., 2023), deep learning (Gao et al., 2020), and ensemble learning (Shen et al., 2020). Some studies focus on practical application scenarios, such as mechanical fault diagnosis (Li et al., 2020), network biology (Theodoris et al., 2023), signal analysis (Wan et al., 2021), physically informed neural networks (Chen et al., 2024), anomaly detection (Ahmad et al., 2024), behavior prediction (Hong et al., 2024), and natural language processing (Li et al., 2024).

2.3. Multi-view learning

MVL aims to utilize data from different sources or with different feature representations and improve the robustness of the model by fusing multi-view information. It has shown its superiority in many tasks such as multi-view dimensionality reduction (Zhang et al., 2025) and multi-view deep learning (Yan et al., 2021).

In clustering tasks, the MVL methods aim to divide data from multiple views into clusters that reflect the potential structure of the data (Yang & Wang, 2018). Based on the approach used to fuse multi-view information, MVC methods can be categorized into subspace learning-based approaches (Kadambari & Chepuri, 2025), graph-based fusion methods (Zhao et al., 2025), non-negative matrix factorization (Liang et al., 2020) and so on. They all have been widely used in text analysis (Fraj et al., 2019), recommendation system (Yu et al., 2018), and bioinformatics (Pfeifer et al., 2023). In recent years, some studies have made breakthroughs in existing methods. Khan et al. (2023a) proposed a novel multi-view subspace clustering method for earning joint representation through low-rank sparse representation. To address the incompleteness problem of MVC, Khan et al. (2024) used weighted concept factorization to solve the problem of missing data instances in each incomplete view. To exploit the manifold structure for consistent representation in the learned kernel space, a novel MVC method is proposed (Khan et al., 2023b), which uses concept factorization to mine the intrinsic information of the data. And it is worth noting that this method avoids the overfitting problem caused by redundant features between views through smoothing the regularization term. Moreover, It uses the correlation constraint to learn the common shared structure from diverse views. Based on this, we focus on the methods used in MVC to ensure efficient view weight allocation and reduce information redundancy. Xu et al. (2025) proposed a novel MVC framework to address the aforementioned two challenges. This framework leverages contrastive learning to enforce consistency across different views. In addition, it enhances the diversity of multi-view representations by ensuring that embedding vectors of samples within a batch are distinct. It also decorrelates the embedding dimensions to further promote representational diversity. Dong et al. (2024) introduced $l_{2,\log}$- (pseudo) norm to achieve the purpose of reducing redundant information, and using pairwise co-regularization to achieve view weighting implicitly.

In classification tasks, MVL also shows significant potential. Based on different fusion mechanisms, it can be divided into co-training based methods (Sevım et al., 2020), late fusion methods (Shi et al., 2022) and so on. Many studies have explored further. Zhuang et al. (2012) improved the classification performance by jointly modeling features from different perspectives and the co-occurrence relationship of documents, Manigrasso et al. (2025) proposed a multi-view deep learning method for mammography classification tasks, and Pintelas et al. (2023) proposed a multi-view convolutional neural network framework for image classification tasks. At present, MVL has been widely used in many classification fields such as natural language processing (Liang et al., 2021), image classification (Wang et al., 2022), object recognition (Zhou et al., 2023).

2.4. MV-LSSVM

MV-LSSVM contains a coupling term that minimizes the error combination of all views to follow the complementarity principle but fails to follow the consistency principle. Given a training set of N data points $\left\{y_k, \mathbf{x}_k^{[v]}\right\}_{k=1}^{k=N}$ for each view $v = 1, \ldots, V$ where $\mathbf{x}_k^{[v]} \in \mathbb{R}^{d^{[v]}}$ and $y_k \in \{-1, 1\}$, the primary formulation of MV-LSSVM model is:

$$\begin{aligned} \min_{\substack{\boldsymbol{w}^{[v]}, \boldsymbol{e}^{[v]}, \\ b^{[v]}}} \quad & \frac{1}{2}\sum_{v=1}^{V} \boldsymbol{w}^{[v]T}\boldsymbol{w}^{[v]} + \frac{1}{2}\sum_{v=1}^{V}\gamma^{[v]}\boldsymbol{e}^{[v]T}\boldsymbol{e}^{[v]} \\ & + \rho \sum_{\substack{v,\mu=1 \\ v\neq\mu}}^{V} \boldsymbol{e}^{[v]T}\boldsymbol{e}^{[\mu]} \\ \text{s.t.} \quad & \boldsymbol{Z}^{[v]T}\boldsymbol{w}^{[v]} + \boldsymbol{y}b^{[v]} = \mathbf{1}_N - \boldsymbol{e}^{[v]}, \quad \text{for } v = 1, \ldots, V, \end{aligned} \tag{1}$$

where $\boldsymbol{e}^{[v]} \in \mathbb{R}^N$ are error variables, $\boldsymbol{y} = (y_1, y_2, \ldots, y_N)$, $\boldsymbol{Z}^{[v]T} \in \mathbb{R}^{N\times d^{[v]}}$ defined as $\boldsymbol{Z}^{[v]T} = [y_1\mathbf{x}_1^{[v]T}, y_2\mathbf{x}_2^{[v]T}, \ldots, y_N\mathbf{x}_N^{[v]T}]$.

This primal optimization function consists of a sum of V different classification objectives (one corresponding to each view), which are coupled through the coupling term $\sum_{\substack{v,\mu=1 \\ v\neq\mu}}^{V} \boldsymbol{e}^{[v]T}\boldsymbol{e}^{[\mu]}$, where ρ is an additional regularization constant, referred to as the coupling parameter. This term aims to minimize the product of the error variables from both views. In this manner, the model incorporates information from both views, allowing a high error variable for a certain sample in one view to be compensated by a corresponding low error variable in another view.

Through the aforementioned approach, MV-LSSVM satisfies the complementarity principle but does not fully exploit the consistency information of multi-view data.

3. Proposed method and solution

Even though the MV-LSSVM has higher operating efficiency than other SVM-based MVL methods, it has some limitations. 1) It does not mine the consistency information of the multi-view data. 2) To achieve optimal generalization performance and reduce estimation bias, leave-one-out cross-validation is frequently employed as a strategy for parameter selection. However, this is undoubtedly time-consuming. 3) It is a shallow structure, which limits its ability to solve complex problems. 4) It fails to make use of previously acquired knowledge, which restricts improvements in generalization. To overcome these limitations, we focus on integrating deep stacking architecture and transfer learning into the learning framework of MV-LSSVM. In the end, we propose a Deep Multi-view Least Squares Support Vector Machine with Consistency and

Table 1
List of main notations.

Symbol	Denotation	Scope
n	Number of training samples	$\mathbb{N}$
m	Number of viewing angles	$\mathbb{N}$
$d^{[v]}$	Dimension of the sample in v-th view	$\mathbb{N}$
$\boldsymbol{w}_{1v}$	Weight vector of the hyperplane in v-th view	$\mathbb{R}^{d^{[v]}}$
b_{1v}	Bias of the hyperplane in v-th view	$\mathbb{R}$
$\boldsymbol{X}_l^{[v]}$	Input dataset in lth layer	$\mathbb{R}^{n\times(d^{[v]}+(l-1))}$
$\boldsymbol{\phi}_{lv}(\cdot)$	Feature map of the vth view	–
θ_{lv}	Weight of the v-th view	$(0,1)$
$(\boldsymbol{K}_{1v})_{(i,j)}$	$\boldsymbol{\phi}_{1v}{}^T(\boldsymbol{x}_i^{[v]})\cdot\boldsymbol{\phi}_{1v}(\boldsymbol{x}_j{}^{[v]})$	$\mathbb{R}^{n\times n}$
$(\boldsymbol{K}_{lv})_{(i,j)}$	$\boldsymbol{\phi}_{lv}{}^T(\boldsymbol{x}_{li}^{[v]})\cdot\boldsymbol{\phi}_{lv}(\boldsymbol{x}_{lj}{}^{[v]})$	$\mathbb{R}^{n\times n}$
u_l	Weight factor in lth layer	$\mathbb{R}$
$y_{i(l-1)}$	Predicted value from the $(l-1)$th layer	$\{+1,-1\}$
$\boldsymbol{y}$	$(y_1,y_2,\ldots,y_n)^T$	$\mathbb{R}^n$
$\boldsymbol{Y}$	$\left[\boldsymbol{y}^\top,\boldsymbol{y}^\top,\ldots,\boldsymbol{y}^\top\right]^\top$	$\mathbb{R}^{mn}$
$\boldsymbol{y}_{(l-1)}$	$\left(y_{1(l-1)},y_{2(l-1)},\cdots,y_{n(l-1)}\right)^T$	$\mathbb{R}^n$
$\boldsymbol{Y}_{(-1)}$	$\left[\boldsymbol{y}_{(-1)}^\top,\boldsymbol{y}_{(-1)}^\top,\cdots,\boldsymbol{y}_{(-1)}^\top\right]^\top$	$\mathbb{R}^{m(n-1)}$
$\{C_{lv}\}_{v=1}^m, d_l, \lambda_l$	Non-negative parameter	$\mathbb{R}$

Complementarity Principles based on Cross-Output Knowledge Transfer (MDCTM) in this section. In the following, we will introduce the model proposal process and solution method in detail.

3.1. Notations and ideology

We use dataset as $S=\left\{\left(\boldsymbol{x}_i^{[1]},\boldsymbol{x}_i^{[2]},\cdots,\boldsymbol{x}_i^{[m]},y_i\right)\right\}_{i=1}^n\in\mathcal{X}\times\mathcal{Y}=\boldsymbol{X}^{[1]}\times\boldsymbol{X}^{[2]}\times\cdots\times\boldsymbol{X}^{[m]}\times\mathbf{y}$, where $\boldsymbol{X}^{[v]}\subseteq R^{d^{[v]}}$, $y_i\in\mathbf{y}=\{+1,-1\}$. For simplification, we have the main notations presented in Table 1.

The original dataset S is used as the input data of the first layer to build the MVLSSVM-2C. Its framework is shown in Fig. 1. Specifically, MVLSSVM-2C satisfies the consistency principle by minimizing the least squares loss of the inconsistency of predictions between different views. At the same time, it combines the weight distribution strategy to adaptively learn and adjust the weight of each perspective to achieve the complementarity principle.

From the second layer upwards, each layer is deep MVLSSVM-2C based on cross-output knowledge transfer. The framework is shown in the Fig. 2. Firstly, MDCTM is built under a deep stacking framework. Different from the first layer, the input to these layers consists of the original dataset S along with the predicted outputs from all preceding layers, which serve as additional features. In this way, the new feature space contains the prediction information from all previous layers, which helps the model to make more comprehensive decisions at each layer, thereby gradually improving the feature representation and effectively separating the manifolds in the data. Secondly, MDCTM assumes that some parameters or hyperparameters are shared from adjacent layers (source domain and target domain). Therefore, we believe the prediction values between adjacent layers will be somewhat similar. Therefore, we try to train a weight parameter that reflects the degree of influence of the prediction values of all previous lower layers on the current layer. The aim of heterogeneous transfer learning is to utilize knowledge from the source domain and related learning tasks to enhance the development of prediction functions in the target domain, a concept that is also demonstrated in this study.

3.2. Problem formulation and solution

3.2.1. MVLSSVM-2C

In the first layer of MDCTM, we construct the MVLSSVM-2C to find m optimal hyperplanes $f_{1v}(\boldsymbol{x})=\left\langle\boldsymbol{w}_{1v},\boldsymbol{\phi}_{1v}\left(\boldsymbol{x}^{[v]}\right)\right\rangle+b_{1v}$. The MVLSSVM-2C extends LS-SVM to MVL scenarios by minimizing the least squares loss of the inconsistency of predictions between different views and adaptively adjusting the weight of each views.

We build the MVLSSVM-2C model as follows

$$
\begin{aligned}
\min_{\substack{\boldsymbol{w}_{11},\ldots,\boldsymbol{w}_{1m},\\ \theta_{11},\ldots,\theta_{1m}}}\quad & \frac{1}{2}\sum_{v=1}^m\theta_{1v}\|\boldsymbol{w}_{1v}\|^2+\sum_{i=1}^n\sum_{v=1}^m\frac{C_{1v}}{2}(\xi_i^{[v]})^2+\frac{\lambda_1}{2}\sum_{v=1}^m\theta_{1v}^2\\
&+d_1\sum_{i=1}^n\sum_{\substack{v,u=1\\ v\neq u}}^m\left((\boldsymbol{w}_{1u}\cdot\boldsymbol{\phi}_{1u}(\boldsymbol{x}_i^{[u]}))-(\boldsymbol{w}_{1v}\cdot\boldsymbol{\phi}_{1v}(\boldsymbol{x}_i^{[v]}))\right)^2\\
\text{s.t.}\quad & y_i=\boldsymbol{w}_{1v}^T\cdot\boldsymbol{\phi}_{1v}(\boldsymbol{x}_i^{[v]})+\xi_i^{[v]},\\
& \theta_{1v}\geq 0,\quad \sum_{v=1}^m\theta_{1v}=1,\\
& i=1,\ldots,n,\quad v=1,\ldots,m,
\end{aligned}
\tag{2}
$$

where $\boldsymbol{\xi}^{[v]}=\left[\xi_1{}^{[v]},\xi_2{}^{[v]}\cdots,\xi_n{}^{[v]}\right]$ is a slack variable about the vth view. We let $\boldsymbol{\phi}_{1v}\left(\boldsymbol{x}_i^{[v]}\right)=\left[\boldsymbol{\phi}_{1v}\left(\boldsymbol{x}_i^{[v]}\right);1\right]$, $\boldsymbol{w}_{1v}^T=\left[\boldsymbol{w}_{1v}{}^T,b_{1v}\right]$ to eliminate the bias terms b_{1v}.

To further illustrate the basic idea of model (2), we give the following explanation.

- To fully capitalize on the complementary information between different perspectives, the model introduces a non-negative weight combination coefficient $\{\theta_{1v}\}_{v=1}^m$, which adaptively learns and adjusts the weight of each view.
- The slack variable means that the hyperplanes $\left(\boldsymbol{w}_{1v}\cdot\boldsymbol{\phi}_{1v}\left(\boldsymbol{x}^{[v]}\right)\right)=1$ and $\left(\boldsymbol{w}_{1v}\cdot\boldsymbol{\phi}_{1v}\left(\boldsymbol{x}^{[v]}\right)\right)=-1$ are as close as possible to the "middle" of the positive and negative sample points of the vth view. The loss of least squares of the slack variable $\sum_{i=1}^n\sum_{v=1}^m\frac{C_{1v}}{2}\left(\xi_i^{[v]}\right)^2$ minimizes the degree of misclassification in each perspective. At the same time,

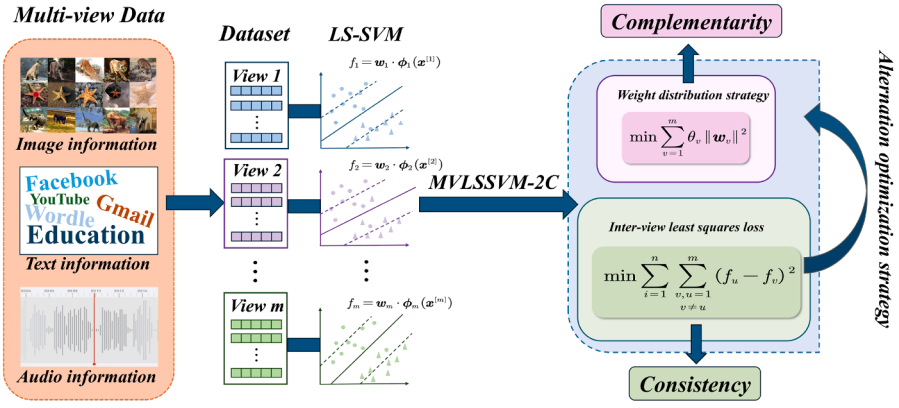


Fig. 1. MVLSSVM-2C algorithm framework.

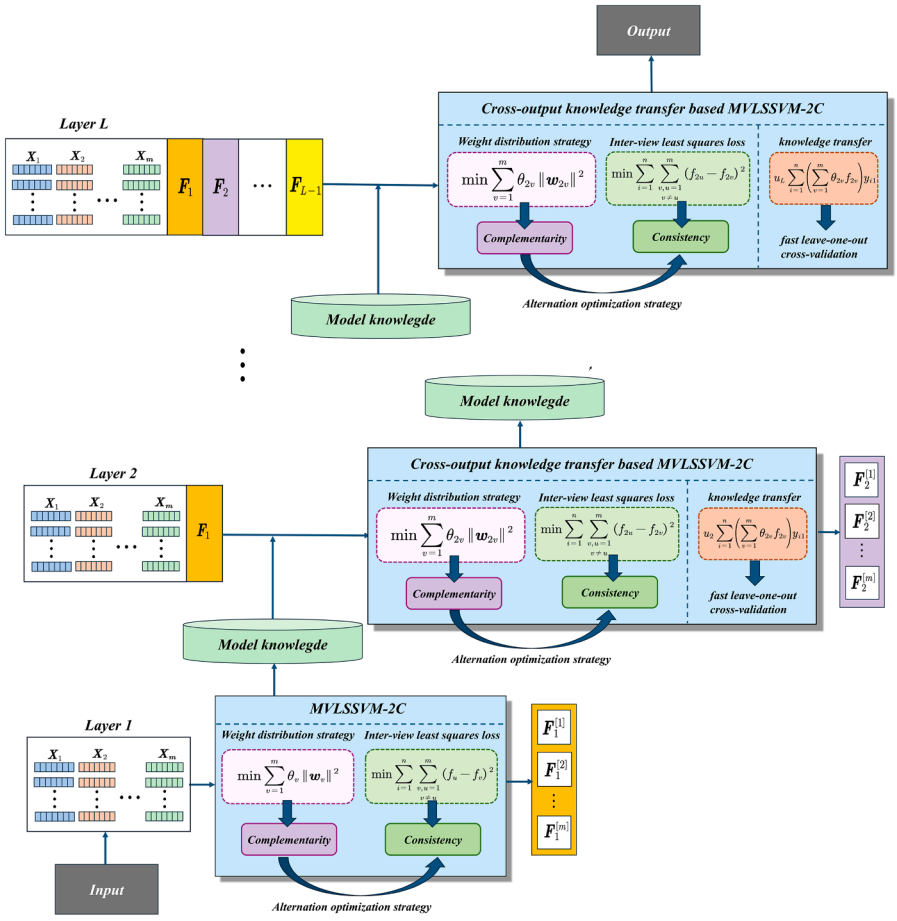


Fig. 2. MDCTM algorithm framework.

the equality constraint $y_i = \boldsymbol{w}_{1v}^T \cdot \boldsymbol{\phi}_{1v}\left(\boldsymbol{x}_i^{[v]}\right) + \xi_i^{[v]}$ simplifies the complexity of the calculation.

- In order to maintain consistency between views, MVLSSVM-2C constructs $\left(\left(\boldsymbol{w}_{1u} \cdot \boldsymbol{\phi}_{1u}\left(\boldsymbol{x}_i^{[u]}\right)\right) - \left(\boldsymbol{w}_{1v} \cdot \boldsymbol{\phi}_{1v}\left(\boldsymbol{x}_i^{[v]}\right)\right)\right)^2$ as the last term of the objective function, which is the least squares loss of the multi-view decision function. In this way, the prediction results of each view for the target sample tend to be consistent.

We use an alternating optimization strategy to solve this optimization problem. Firstly, given $\{\theta_{1v}\}_{v=1}^m$, iterate $\{\boldsymbol{w}_{1v}\}_{v=1}^m$. The optimization problem (2) can be reformulated as:

$$\begin{aligned} \min_{\boldsymbol{w}_{11},\ldots,\boldsymbol{w}_{1m}} \quad & \frac{1}{2}\sum_{v=1}^{m}\theta_{1v}\|\boldsymbol{w}_{1v}\|^2 + \sum_{v=1}^{m}\frac{C_{1v}}{2}(\boldsymbol{\xi}^{[v]})^T\boldsymbol{\xi}^{[v]} \\ & - 2d_1\sum_{\substack{v,u=1 \\ v\neq u}}^{m}(\boldsymbol{\xi}^{[u]})^T\boldsymbol{\xi}^{[v]} \\ \text{s.t.} \quad & \boldsymbol{y} = \boldsymbol{\phi}_{1v}^T(\boldsymbol{X}^{[v]})\boldsymbol{w}_{1v} + \boldsymbol{\xi}^{[v]}, \\ & v = 1,\ldots,m, \end{aligned} \tag{3}$$

where $\boldsymbol{\phi}_{1v}(\boldsymbol{X}^{[v]}) = \left[\boldsymbol{\phi}_{1v}(\boldsymbol{x}_1^{[v]}), \boldsymbol{\phi}_{1v}(\boldsymbol{x}_2^{[v]}), \ldots, \boldsymbol{\phi}_{1v}(\boldsymbol{x}_n^{[v]})\right]$.

By introducing Lagrange multipliers $\boldsymbol{\alpha}_i = (\alpha_{i1}, \alpha_{i2}, \ldots, \alpha_{in})^T$, $i = 1,\ldots,m$. We write the linear equation equivalent to optimization problem (3)

$$(\boldsymbol{H_1} + \boldsymbol{U_1})\boldsymbol{\pi} = \boldsymbol{Y}(\boldsymbol{E}_{mn} - 2d_1(m-1)\boldsymbol{U}_1), \tag{4}$$

where

$$\boldsymbol{H}_1 = \begin{bmatrix} \frac{1}{\theta_{11}}\boldsymbol{K}_{11} & -\frac{2d_1}{\theta_{12}C_{11}}\boldsymbol{K}_{12} & \cdots & -\frac{2d_1}{\theta_{1m}C_{11}}\boldsymbol{K}_{1m} \\ -\frac{2d_1}{\theta_{11}C_{12}}\boldsymbol{K}_{11} & \frac{1}{\theta_{12}}\boldsymbol{K}_{12} & \cdots & -\frac{2d_1}{\theta_{1m}C_{12}}\boldsymbol{K}_{1m} \\ \vdots & \vdots & \ddots & \vdots \\ -\frac{2d_1}{\theta_{11}C_{1m}}\boldsymbol{K}_{11} & -\frac{2d_1}{\theta_{12}C_{1m}}\boldsymbol{K}_{12} & \cdots & \frac{1}{\theta_{1m}}\boldsymbol{K}_{1m} \end{bmatrix}, \tag{5}$$

$$\boldsymbol{U}_1 = \begin{bmatrix} \frac{1}{C_{11}}\boldsymbol{E}_n & \boldsymbol{O}_n & \cdots & \boldsymbol{O}_n \\ \boldsymbol{O}_n & \frac{1}{C_{12}}\boldsymbol{E}_n & \cdots & \boldsymbol{O}_n \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{O}_n & \boldsymbol{O}_n & \cdots & \frac{1}{C_{1m}}\boldsymbol{E}_n \end{bmatrix}, \tag{6}$$

$$\boldsymbol{\pi} = \left[\boldsymbol{\alpha}_1^T, \boldsymbol{\alpha}_2^T, \ldots, \boldsymbol{\alpha}_m^T\right]^T. \tag{7}$$

By solving the Eq. (4), we can obtain the optimal $\boldsymbol{\pi}$, namely $\left[\boldsymbol{\alpha}_1^*, \boldsymbol{\alpha}_2^*, \ldots, \boldsymbol{\alpha}_m^*\right]^T$.

Secondly, given $\{\boldsymbol{w}_{1v}\}_{v=1}^m$, iterate $\{\theta_{1v}\}_{v=1}^m$. For a given $\{\boldsymbol{w}_{1v}\}_{v=1}^m$, the optimization problem (2) can be expressed as:

$$\begin{aligned} \min_{\theta_{11},\ldots,\theta_{1m}} \quad & \frac{1}{2}\sum_{v=1}^{m}\theta_{1v}\|\boldsymbol{w}_{1v}\|^2 + \frac{\lambda_1}{2}\sum_{v=1}^{m}\theta_{1v}^2 \\ \text{s.t.} \quad & \sum_{v=1}^{m}\theta_{1v} = 1, \quad \theta_{1v} \geq 0, \quad v = 1,\ldots,m, \end{aligned} \tag{8}$$

this is a convex optimization problem involving quadratic functions and we can get optimal $\{\theta_{1v}^*\}_{v=1}^m$.

After the first layer, the original input $\boldsymbol{x}_i$ can be classified via the decision function

$$f_1(\boldsymbol{x}_i) = \operatorname{sign}\left(\sum_{v=1}^{m} \frac{\theta_{1v}^*}{\theta_{1v}} \boldsymbol{\alpha}_v^{*T} \boldsymbol{\phi}_{1v}^T(\boldsymbol{X}^{[v]}) \boldsymbol{\phi}_{1v}(\boldsymbol{x}_i^{[v]})\right), \tag{9}$$

where $\boldsymbol{x}_{1i}^{[v]}$ denotes the original input $\boldsymbol{x}_i^{[v]}$ in the first layer, then we can get the prediction value of the first layer written as $\boldsymbol{F}_1 = \left(f_1(\boldsymbol{x}_1), f_1(\boldsymbol{x}_2), \cdots, f_1(\boldsymbol{x}_n)\right)^T$.

3.2.2. Deep MVLSSVM-2C based on cross-output knowledge transfer

We regard the input dataset of the $(l-1)$th $(l = 2, \cdots, L)$ layer as the source domain $D_{S(l-1)}$ and the input dataset of the lth layer as the target domain D_{Tl}. Starting from the second layer, the input dataset $\boldsymbol{X}_l^{[v]}$ of the model is formed by splicing the predicted value $\boldsymbol{F}_{l-1}$ of the $(l-1)$th layer as the last attribute to the input dataset $\boldsymbol{X}_{l-1}^{[v]}$ of the $(l-1)$th layer, which can be defined as $\boldsymbol{X}_{l-1}^{[v]} \bigoplus \boldsymbol{F}_{l-1}$. Therefore, the new input dataset $\boldsymbol{X}_l^{[v]}$ contains the prediction information of the $(l-1)$th layer, which helps to separate the manifold in the original data, thereby achieving better separability and improving the generalization ability of the current layer. Using the relevant knowledge of parameter-based transfer learning, we assume that the outputs of adjacent layers will have certain similarities. In other words, we aim to maximize $\sum_{i=1}^{n} f_l(x_i) y_{i(l-1)}$. A weighting factor u_l is utilized to reflect the degree of influence that the learned output knowledge from $D_{S(l-1)}$ exerts on D_{Tl}. The implementation method of the 2C principles is the same as MVLSSVM-2C. Accordingly, the optimization problem changes to

$$\begin{aligned}
\min_{\substack{\boldsymbol{w}_{l1},\ldots,\boldsymbol{w}_{lm},\\ \theta_{l1},\ldots,\theta_{lm}}} \quad & \frac{1}{2}\sum_{v=1}^{m}\theta_{lv}\|\boldsymbol{w}_{lv}\|^2 + \sum_{i=1}^{n}\sum_{v=1}^{m}\frac{C_{lv}}{2}(\xi_{li}^{[v]})^2 \\
& - \mu_l \sum_{i=1}^{n}\left(\sum_{v=1}^{m}\theta_{lv}(y_i - \xi_{li}^{[v]})\right) y_{i(l-1)} \\
& + d_l \sum_{i=1}^{n}\sum_{\substack{v,u=1\\ v\neq u}}^{m}\left((y_i - \xi_{li}^{[u]}) - (y_i - \xi_{li}^{[v]})\right)^2 \\
& + \frac{\lambda_l}{2}\sum_{v=1}^{m}\theta_{lv}^2 \\
\text{s.t.} \quad & y_i = \boldsymbol{w}_{lv}^T \boldsymbol{\phi}_v(\boldsymbol{x}_{li}^{[v]}) + \xi_{li}^{[v]}, \\
& \sum_{v=1}^{m}\theta_{lv} = 1, \quad \theta_{lv} \geq 0, \\
& i = 1, \ldots, n, \quad v = 1, \ldots, m,
\end{aligned} \tag{10}$$

where $\xi_l^{[v]} = \left[\xi_{l1}^{[v]}, \xi_{l2}^{[v]}, \cdots, \xi_{ln}^{[v]}\right]^T$ is the slack variable of the lth layer with respect to the vth view. We provide the following illustrations of the MDCTM.

- The model adopts a deep stacking architecture, which can enhance ability to handle complex problems. Starting from the second layer, the input data $\boldsymbol{X}_l^{[v]}$ of each layer uses the predicted outputs of all previous layers, effectively revealing the manifold structure of the original data.
- Each layer adheres to the 2C principles. This ensures that each layer fully exploits the consistency and complementary information of multi-view data. The least squares loss $\sum_{i=1}^{n}\sum_{\substack{v,u=1\\ v\neq u}}^{m}\left(\left(y_i - \xi_{li}^{[u]}\right) - \left(y_i - \xi_{li}^{[v]}\right)\right)^2$ enforces consistency, while adaptive view weighting $\{\theta_{lv}\}_{v=1}^{m}$ ensures complementarity. The model is optimized via an alternating optimization strategy.
- Following transfer learning, each layer learns a weight parameter μ_l to quantify the influence of preceding layers' predictions, which can improve the classification performance of the model. This parameter is efficiently estimated via fast leave-one-out cross-validation, enabling adaptive and effective inter-layer knowledge transfer.
- MDCTM permits random allocation of parameters (e.g., weights and kernel widths) within each layer, enhancing computational efficiency.

We continue to use the alternating optimization strategy. Firstly, for the given $\{\theta_{lv}\}_{v=1}^{m}$, update $\{\boldsymbol{w}_{lv}\}_{v=1}^{m}$. The optimization problem (10) can be expressed as

$$\begin{aligned}
\min_{\boldsymbol{w}_{l1},\ldots,\boldsymbol{w}_{lm}} \quad & \frac{1}{2}\sum_{v=1}^{m}\theta_{lv}\|\boldsymbol{w}_{lv}\|^2 + \sum_{i=1}^{n}\sum_{v=1}^{m}\frac{C_{lv}}{2}(\xi_{li}^{[v]})^2 \\
& - u_l \sum_{i=1}^{n}\left(y_i - \sum_{v=1}^{m}\theta_{lv}\xi_{li}^{[v]}\right) y_{i(l-1)} \\
& - 2d_l \sum_{i=1}^{n}\sum_{\substack{v,u=1\\ v\neq u}}^{m}\xi_{li}^{[u]}\xi_{li}^{[v]} \\
\text{s.t.} \quad & y_i = \boldsymbol{w}_{lv}^T \boldsymbol{\phi}_v(\boldsymbol{x}_{li}^{[v]}) + \xi_{li}^{[v]}, \\
& i = 1, \ldots, n, \quad v = 1, \ldots, m.
\end{aligned} \tag{11}$$

The Lagrangian function in vector form as

$$\begin{aligned}
L = & \frac{1}{2}\sum_{v=1}^{m}\theta_{lv}\|\boldsymbol{w}_{lv}\|^2 + \sum_{v=1}^{m}\frac{C_v}{2}(\boldsymbol{\xi}_l^{[v]})^T \boldsymbol{\xi}_l^{[v]} \\
& - u_l \boldsymbol{y}_{(l-1)}^T \left(\boldsymbol{y} - \sum_{v=1}^{m}\theta_{lv}\boldsymbol{\xi}_l^{[v]}\right) - 2d_l \sum_{\substack{v,u=1\\ v\neq u}}^{m}(\boldsymbol{\xi}_l^{[u]})^T \boldsymbol{\xi}_l^{[v]} \\
& + \sum_{v=1}^{m}\boldsymbol{\alpha}_{lv}^T \cdot \left(\boldsymbol{y} - \boldsymbol{\phi}_v^T(\boldsymbol{X}_l^{[v]}) \cdot \boldsymbol{w}_{lv} - \boldsymbol{\xi}_l^{[v]}\right),
\end{aligned} \tag{12}$$

where $\boldsymbol{\phi}_v(\mathbf{X}_l^{[v]}) = \left[\boldsymbol{\phi}_v(\boldsymbol{x}_{l1}^{[v]}), \boldsymbol{\phi}_v(\boldsymbol{x}_{l2}^{[v]}), \ldots, \boldsymbol{\phi}_v(\boldsymbol{x}_{ln}{}^{[v]})\right]$, the Lagrange multipliers are $\boldsymbol{\alpha}_{lv} = \left(\alpha_{l1}{}^{[v]}, \alpha_{l2}{}^{[v]}, \cdots, \alpha_{ln}{}^{[v]}\right)^T$, $v = 1, \ldots, m$. We can derive the following conditions known as Karush-Kuhn-Tucker (KKT):

$$\frac{\partial L}{\partial \boldsymbol{w}_{lv}} = \theta_{lv}\boldsymbol{w}_v - \boldsymbol{\phi}_v(\boldsymbol{X}_l^{[v]})\boldsymbol{\alpha}_{lv} = 0, \tag{13}$$

$$\frac{\partial L}{\partial \boldsymbol{\xi}_l^{[v]}} = C_{lv}\boldsymbol{\xi}_l^{[v]} + \theta_{lv}u_l\boldsymbol{y}_{(l-1)} - 2d_l\sum_{\substack{v,u=1\\ v\neq u}}^{m}\boldsymbol{\xi}_l^{[v]} - \boldsymbol{\alpha}_{lv} = 0, \tag{14}$$

$$\boldsymbol{y} - \boldsymbol{\phi}_v^T(\boldsymbol{X}_l^{[v]})\boldsymbol{w}_{lv} - \boldsymbol{\xi}_l^{[v]} = 0. \tag{15}$$

By setting the coefficient matrix of the slack variable $\xi_l^{[v]}$ to be $\boldsymbol{A}_l$, we can combine all the views in Eq. (14), and therefore obtain $\boldsymbol{A}_l^{-1} = |\boldsymbol{A}_l|^{-1}\boldsymbol{A}_l^*$, $\boldsymbol{A}_l^*$ is the adjoint matrix, where

$$\boldsymbol{A}_l = \begin{pmatrix} C_{l1} & -2d_l & \cdots & -2d_l \\ -2d_l & C_{l2} & \cdots & -2d_l \\ \vdots & \vdots & \ddots & \vdots \\ -2d_l & -2d_l & \cdots & C_{lm} \end{pmatrix}, \boldsymbol{A}_l^* = \begin{pmatrix} A_{11l} & A_{21l} & \cdots & A_{m1l} \\ A_{12l} & A_{22l} & \cdots & A_{m2l} \\ \vdots & \vdots & \ddots & \vdots \\ A_{1ml} & A_{2ml} & \cdots & A_{mml} \end{pmatrix}, \tag{16}$$

after that, we have

$$\begin{bmatrix} \xi_l^{[1]} \\ \xi_l^{[2]} \\ \vdots \\ \xi_l^{[m]} \end{bmatrix} = \frac{1}{|\boldsymbol{A}_l|}\begin{bmatrix} \sum_{v=1}^{m} A_{v1l}\left(\boldsymbol{\alpha}_{lv} - \theta_{l1}u_l\boldsymbol{y}_{(l-1)}\right) \\ \sum_{v=1}^{m} A_{v2l}\left(\boldsymbol{\alpha}_{lv} - \theta_{l2}u_l\boldsymbol{y}_{(l-1)}\right) \\ \vdots \\ \sum_{v=1}^{m} A_{vml}\left(\boldsymbol{\alpha}_{lv} - \theta_{lm}u_l\boldsymbol{y}_{(l-1)}\right) \end{bmatrix}. \tag{17}$$

Combining Eqs. (13), (15) and (17), we can get equivalent linear equations to the optimization problem (11)

$$\left(\boldsymbol{H}_l + \frac{1}{|\boldsymbol{A}_l|}\boldsymbol{U}_l\right)\boldsymbol{\pi}_l = \boldsymbol{Y} - \frac{u_l}{|\boldsymbol{A}_l|}\boldsymbol{R}_l, \tag{18}$$

where

$$\boldsymbol{H}_l = \begin{bmatrix} \frac{1}{\theta_{l1}}\boldsymbol{K}_{l1} & \boldsymbol{O}_n & \cdots & \boldsymbol{O}_n \\ \boldsymbol{O}_n & \frac{1}{\theta_{l2}}\boldsymbol{K}_{l2} & \cdots & \boldsymbol{O}_n \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{O}_n & \boldsymbol{O}_n & \cdots & \frac{1}{\theta_{lm}}\boldsymbol{K}_{lm} \end{bmatrix}, \tag{19}$$

$$\boldsymbol{U}_l = \begin{bmatrix} A_{11l}\boldsymbol{E}_n & A_{21l}\boldsymbol{E}_n & \cdots & A_{m1l}\boldsymbol{E}_n \\ A_{12l}\boldsymbol{E}_n & A_{22l}\boldsymbol{E}_n & \cdots & A_{m2l}\boldsymbol{E}_n \\ \vdots & \vdots & \ddots & \vdots \\ A_{1ml}\boldsymbol{E}_n & A_{2ml}\boldsymbol{E}_n & \cdots & A_{mml}\boldsymbol{E}_n \end{bmatrix}, \tag{20}$$

$$\boldsymbol{\pi}_l = \left[\boldsymbol{\alpha}_{l1}^T, \boldsymbol{\alpha}_{l2}^T, \ldots, \boldsymbol{\alpha}_{lm}^T\right]^T, \tag{21}$$

$$\boldsymbol{R}_l = \left[\sum_{v=1}^{m}\theta_{l1}A_{v1l}\boldsymbol{y}_{(l-1)}^T, \sum_{v=1}^{m}\theta_{l2}A_{v2l}\boldsymbol{y}_{(l-1)}^T, \ldots, \sum_{v=1}^{m}\theta_{lm}A_{vml}\boldsymbol{y}_{(l-1)}^T\right]^T. \tag{22}$$

By solving the Eq. (18), we can obtain the optimal $\boldsymbol{\pi}_l$, namely $\left[\boldsymbol{\alpha}_{l1}^*, \boldsymbol{\alpha}_{l2}^*, \ldots, \boldsymbol{\alpha}_{lm}^*\right]^T$.

Secondly, we use the obtained $\left\{\boldsymbol{w}_{lv}^*\right\}_{v=1}^m$ to update $\left\{\theta_{lv}\right\}_{v=1}^m$. The optimization problem (10) can be written as

$$\begin{aligned} \min_{\theta_{l1},\theta_{l2},\ldots,\theta_{lm}} \quad & \frac{1}{2}\sum_{v=1}^{m}\theta_{lv}\|\boldsymbol{w}_{lv}^*\|^2 + \frac{\lambda_l}{2}\sum_{v=1}^{m}\theta_{lv}^2 \\ & - u_l\sum_{i=1}^{n}\left(\sum_{v=1}^{m}\theta_{lv}\boldsymbol{w}_{lv}^T\boldsymbol{\phi}_v\left(\boldsymbol{x}_{li}^{[v]}\right)\right)y_{i(l-1)} \\ \text{s.t.} \quad & \sum_{v=1}^{m}\theta_{lv} = 1, \quad \theta_{lv} \geq 0, \\ & v = 1, 2, \ldots, m, \end{aligned} \tag{23}$$

this also constitutes a convex quadratic optimization problem from which we can derive optimal $\left\{\theta_{lv}^*\right\}_{v=1}^m$. We can get the decision function

$$f_l(\boldsymbol{x}_i) = \text{sign}\left(\sum_{v=1}^{m}\frac{\theta_{lv}^*}{\theta_{lv}}\boldsymbol{\alpha}_{lv}^{*T}\boldsymbol{\phi}_v^T(\boldsymbol{X}_l^{[v]})\boldsymbol{\phi}(\boldsymbol{x}_{li}^{[v]})\right), \tag{24}$$

and $\boldsymbol{F}_l = \left(f_l(\boldsymbol{x}_1), f_l(\boldsymbol{x}_2), \cdots, f_l(\boldsymbol{x}_n)\right)^T$ for $\boldsymbol{X}_l$. Additional processing layers are incorporated until the prediction accuracy no longer improves or the improvements become insignificant, i.e., $\left\|\boldsymbol{F}_l - \boldsymbol{F}_{l-1}\right\|_F^2 < \varepsilon$.

3.3. Parameter optimization

To efficiently find the optimal parameter value u_l in the optimization problem (10) to determine the degree of cross-output knowledge transfer between consecutive modules, for optimizing parameters, we introduce a fast leave-one-out cross-validation method. we let $\boldsymbol{P}_l = \boldsymbol{H}_l + \frac{1}{|\boldsymbol{A}_l|}\boldsymbol{U}_l$ in Eq. (18) and $\boldsymbol{Q}_l = \boldsymbol{P}_l^{-1}$. Taking the first iteration as an example, excluding the first training sample, that is, using the $2 \sim n$ samples to train the model. We separate the $\boldsymbol{P}_l$ into blocks according to Eqs. (19) and (20) to obtain the first row and first column of each block matrix.

$$\left[\begin{array}{cc|cc|c|cc} h_{11l} & \boldsymbol{h}_{1l}^T & f_{11l}^{21} & \boldsymbol{f}_{21l}^T & \cdots & f_{11l}^{m1} & \boldsymbol{f}_{m1l}^T \\ \boldsymbol{h}_{1l} & \boldsymbol{H}_{(-1)l}^1 & \boldsymbol{f}_{21l} & \boldsymbol{F}_{(-1)l}^{21} & \cdots & \boldsymbol{f}_{m1l} & \boldsymbol{F}_{(-1)l}^{m1} \\ \hline f_{11l}^{12} & \boldsymbol{f}_{12l}^T & h_{22l} & \boldsymbol{h}_{2l}^T & \cdots & f_{11l}^{m2} & \boldsymbol{f}_{m2l}^T \\ \boldsymbol{f}_{12l} & \boldsymbol{F}_{(-1)l}^{12} & \boldsymbol{h}_{2l} & \boldsymbol{H}_{(-1)l}^2 & \cdots & \boldsymbol{f}_{m2l} & \boldsymbol{F}_{(-1)l}^{m2} \\ \hline \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \hline f_{11l}^{1m} & \boldsymbol{f}_{1ml}^T & f_{11l}^{2m} & \boldsymbol{f}_{2ml}^T & \cdots & h_{mml} & \boldsymbol{h}_{ml}^T \\ \boldsymbol{f}_{1ml} & \boldsymbol{F}_{(-1)ml}^{1m} & \boldsymbol{f}_{2ml} & \boldsymbol{F}_{(-1)l}^{2m} & \cdots & \boldsymbol{h}_{ml} & \boldsymbol{H}_{(-1)l}^m \end{array}\right] \tag{25}$$

Then the predicted value of the first sample can be completely given by $\boldsymbol{\beta}_{l1} = \left(\alpha_{l1}^{[1]}, \alpha_{l1}^{[2]}, \cdots, \alpha_{l1}^{[m]}\right)^T$, i.e., the Lagrange multiplier corresponding to the first sample. We can obtain the predicted value of the first sample as follows

$$\tilde{y}_{1l} = \boldsymbol{G}_{11l}\boldsymbol{\beta}_{l1} + y_1 - \frac{u_l}{|\boldsymbol{A}_l|}\sum_{v=1}^{m}\sum_{u=1}^{m}\theta_{lv}A_{uvl}y_{1(l-1)}, \tag{26}$$

where

$$\begin{aligned} \boldsymbol{G}_{11l} = {} & \theta_{l1}\left[\boldsymbol{h}_{1l}^T, \boldsymbol{f}_{21l}^T, \cdots, \boldsymbol{f}_{m1l}^T\right]\boldsymbol{Q}_{(-1)l}\boldsymbol{F}_l \\ & + \theta_{l2}\left[\boldsymbol{f}_{12l}^T, \boldsymbol{h}_{2l}^T, \cdots, \boldsymbol{f}_{m2l}^T\right]\boldsymbol{Q}_{(-1)l}\boldsymbol{F}_l \\ & + \cdots \\ & + \theta_{l1}\left[\boldsymbol{f}_{1ml}^T, \boldsymbol{f}_{2ml}^T, \cdots, \boldsymbol{h}_{ml}^T\right]\boldsymbol{Q}_{(-1)l}\boldsymbol{F}_l \\ & - \theta_{l1}\left(h_{11l}, f_{11l}^{21}, \cdots, f_{11l}^{m1}\right) \\ & - \theta_{l2}\left(f_{11l}^{12}, h_{22l}, \cdots, f_{11l}^{m2}\right) \\ & - \cdots \\ & - \theta_{lm}\left(f_{11l}^{1m}, f_{11l}^{2m}, \cdots, h_{mml}\right), \end{aligned} \tag{27}$$

$$\boldsymbol{P}_{(-1)l} = \begin{bmatrix} \boldsymbol{H}_{(-1)l}^1 & \boldsymbol{F}_{(-1)l}^{21} & \cdots & \boldsymbol{F}_{(-1)l}^{m1} \\ \boldsymbol{F}_{(-1)l}^{12} & \boldsymbol{H}_{(-1)l}^2 & \cdots & \boldsymbol{F}_{(-1)l}^{m2} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{F}_{(-1)l}^{1m} & \boldsymbol{F}_{(-1)l}^{2m} & \cdots & \boldsymbol{H}_{(-1)l}^m \end{bmatrix}, \tag{28}$$

$$\boldsymbol{Q}_{(-1)l} = \boldsymbol{P}_{(-1)l}^{-1}, \tag{29}$$

$$\boldsymbol{C}_l = \begin{bmatrix} \boldsymbol{h}_{1l} & \boldsymbol{f}_{21l} & \cdots & \boldsymbol{f}_{m1l} \\ \boldsymbol{f}_{12l} & \boldsymbol{h}_{2l} & \cdots & \boldsymbol{f}_{m2l} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{f}_{1ml} & \boldsymbol{f}_{2ml} & \cdots & \boldsymbol{h}_{ml} \end{bmatrix}. \tag{30}$$

Proof. To solve this problem, we introduce Lagrange multiplier $\boldsymbol{\alpha}_{(-1)lv} = \left(\alpha_{(-1)l2}^{[v]}, \alpha_{(-1)l3}^{[v]}, \ldots, \alpha_{(-1)ln}^{[v]}\right)^T$, we use the $2 \sim n$ samples to train the model parameters

$$\boldsymbol{P}_{(-1)l}\boldsymbol{\pi}_{(-1)l} = \boldsymbol{Y}_{(-1)} - \frac{u_l}{|\boldsymbol{A}_l|}\boldsymbol{R}_{(-1)l}, \tag{31}$$

where

$$\boldsymbol{\pi}_{(-1)l} = \left[\boldsymbol{\alpha}_{(-1)l1}^T, \boldsymbol{\alpha}_{(-1)l2}^T, \ldots, \boldsymbol{\alpha}_{(-1)lm}^T\right]^T, \tag{32}$$

$$\boldsymbol{R}_{(-1)l} = \begin{bmatrix} \sum_{v=1}^{m}\theta_{l1}A_{v1l}\boldsymbol{y}_{(l-1)(-1)} \\ \sum_{v=1}^{m}\theta_{l2}A_{v2l}\boldsymbol{y}_{(l-1)(-1)} \\ \vdots \\ \sum_{v=1}^{m}\theta_{lm}A_{vml}\boldsymbol{y}_{(l-1)(-1)} \end{bmatrix}, \tag{33}$$

where $\boldsymbol{y}_{(l-1)(-1)} = \left(y_{2(l-1)}, y_{3(l-1)}, \ldots, y_{n(l-1)}\right)^T$ and $\boldsymbol{y}_{(-1)} = \left(y_2, y_3, \ldots, y_n\right)^T$.

The first training sample is assigned the predicted label

$$\begin{aligned} \tilde{y}_{1l} = {} & \sum_{v=1}^{m}\theta_{lv}\boldsymbol{\alpha}_{(-1)lv}^T\boldsymbol{\phi}_v^T\left(\boldsymbol{X}_{(-1)l}^{[v]}\right)\boldsymbol{\phi}_v\left(\boldsymbol{x}_{li}^{[v]}\right) \\ = {} & \theta_{l1}\left[\boldsymbol{h}_{1l}^T, \boldsymbol{f}_{21l}^T, \ldots, \boldsymbol{f}_{m1l}^T\right]\boldsymbol{\pi}_{(-1)l} \\ & + \theta_{l2}\left[\boldsymbol{f}_{12l}^T, \boldsymbol{h}_{2l}^T, \ldots, \boldsymbol{f}_{m2l}^T\right]\boldsymbol{\pi}_{(-1)l} \\ & + \cdots \\ & + \theta_{lm}\left[\boldsymbol{f}_{1ml}^T, \boldsymbol{f}_{2ml}^T, \ldots, \boldsymbol{h}_{ml}^T\right]\boldsymbol{\pi}_{(-1)l}, \end{aligned} \tag{34}$$

where $\boldsymbol{\phi}_v\left(\boldsymbol{X}^{[v]}_{(-1)l}\right) = \left[\boldsymbol{\phi}_v\left(\boldsymbol{x}^{[v]}_{l2}\right), \boldsymbol{\phi}_v\left(\boldsymbol{x}^{[v]}_{l3}\right), \ldots, \boldsymbol{\phi}_v\left(\boldsymbol{x}^{[v]}_{ln}\right)\right]$. Considering the $2 \sim n$, $n+2 \sim 2n$, $\cdots$, $(m-1)n+2 \sim mn$ equations in Eq. (18) after partitioning, we have

$$\boldsymbol{C}_l\boldsymbol{\beta}_{l1} + \boldsymbol{P}_{(-1)l}\boldsymbol{\pi}_{(-1)l} = \boldsymbol{Y}_{(-1)} - \frac{u_l}{|\boldsymbol{A}_l|}\boldsymbol{R}_{(-1)l}. \tag{35}$$

Similarly, we consider the $1, n+1, \cdots, (m-1)n+1$ equations

$$\boldsymbol{L}_l\boldsymbol{\beta}_{l1} + \boldsymbol{C}'_l\boldsymbol{\pi}_{(-1)l} = \boldsymbol{y}^1 - \frac{u_l}{|\boldsymbol{A}_l|}\boldsymbol{R}^1_l, \tag{36}$$

$$\boldsymbol{L}_l = \begin{bmatrix} h_{11l} & f^{21}_{11l} & \cdots & f^{m1}_{11l} \\ f^{12}_{11l} & h_{22l} & \cdots & f^{m2}_{11l} \\ \vdots & \vdots & \ddots & \vdots \\ f^{1m}_{11l} & f^{2m}_{11l} & \cdots & h_{mml} \end{bmatrix}, \tag{37}$$

$$\boldsymbol{C}'_l = \begin{bmatrix} \boldsymbol{h}^T_{1l} & \boldsymbol{f}^T_{21l} & \cdots & \boldsymbol{f}^T_{m1l} \\ \boldsymbol{f}^T_{12l} & \boldsymbol{h}^T_{2l} & \cdots & \boldsymbol{f}^T_{m2l} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{f}^T_{1ml} & \boldsymbol{f}^T_{2ml} & \cdots & \boldsymbol{h}^T_{ml} \end{bmatrix}, \tag{38}$$

$$\boldsymbol{y}^1 = \left(y_1, y_1, \cdots, y_1\right)^T, \tag{39}$$

$$\boldsymbol{R}^1_l = \begin{pmatrix} \sum_{v=1}^{m} \theta_{l1} A_{v1l} y_{1(l-1)} \\ \sum_{v=1}^{m} \theta_{l2} A_{v2l} y_{1(l-1)} \\ \vdots \\ \sum_{v=1}^{m} \theta_{lm} A_{vml} y_{1(l-1)} \end{pmatrix}. \tag{40}$$

By substituting Eqs. (31), (35), and (36) into Eq. (34) in turn, we can obtain Eq. (26). □

Since the linear equation system (18) is not sensitive to the order of the equations, so for the prediction case excluding the ith sample, we can obtain it similarly by exchanging the position of the block matrix in Eq. (18), then we express it as

$$\tilde{y}_{il} = \boldsymbol{G}_{iil}\boldsymbol{\beta}_{li} + y_i - \frac{u_l}{|\boldsymbol{A}_l|}\sum_{v=1}^{m}\sum_{u=1}^{m}\theta_{lv}A_{uvl}y_{i(l-1)}. \tag{41}$$

After that, we set $\boldsymbol{\alpha}'_l = \boldsymbol{Q}_l\boldsymbol{y}$ and $\boldsymbol{\alpha}''_l = \frac{1}{|\boldsymbol{A}_l|}\boldsymbol{Q}_l\boldsymbol{R}_l$, and substitute into Eq. (41) we have

$$\tilde{y}_{il} = y_i + \boldsymbol{G}_{iil}\boldsymbol{\gamma}'_{il} - u_l\boldsymbol{G}_{iil}\boldsymbol{\gamma}''_{il} - \frac{u_l}{|\boldsymbol{A}_l|}\sum_{v=1}^{m}\sum_{u=1}^{m}\theta_{lv}A_{uvl}y_{i(l-1)}, \tag{42}$$

where $\boldsymbol{\gamma}'_{il} = \left(\alpha'_{il}, \alpha'_{(i+n)l}, \cdots, \alpha'_{(m-1)nl}\right)^T$ and $\boldsymbol{\gamma}''_{il} = \left(\alpha''_{il}, \alpha''_{(i+n)l}, \cdots, \alpha''_{(m-1)nl}\right)^T$. The optimal u_l should maintain the same sign as $\tilde{y}_{il}$ and y_i across all training samples. However, the non-convex formulation may lead to numerous local minima issues. Therefore, we employ the following loss function

$$\begin{aligned} l\left(\tilde{y}_{il}, y_i\right) &= \left|1 - \tilde{y}_{il}y_i\right|_+ \\ &= \left|u_l\boldsymbol{G}_{iil}\boldsymbol{\gamma}''_{il} - \boldsymbol{G}_{iil}\boldsymbol{\gamma}'_{il} + \frac{u_l}{m|\boldsymbol{A}_l|}\sum_{v=1}^{m}\sum_{u=1}^{m}A_{uvl}y_{i(l-1)}\right|_+, \end{aligned} \tag{43}$$

which resembles the hinge loss, where $|\boldsymbol{x}|_+ = \max\{0, \boldsymbol{x}\}$. This provides a convex upper bound on the leave-one-out misclassification loss and favors solutions where $\tilde{y}_{il}$ has an absolute value of at least 1 and shares the same sign as y_i. Finally, the optimization problem is

$$\begin{aligned} \min_{u_l} \quad & \sum_{i=1}^{n} l\left(\tilde{y}_{il}, y_i\right) \\ \text{s.t.} \quad & 0 \le u_l \le B, \end{aligned} \tag{44}$$

where B is a constant. Regularization based on this can enhance numerical stability. The optimization process can be carried out using a projected sub-gradient descent algorithm.

3.4. Algorithms

After the theoretical part is proposed, the MDCTM algorithm is presented in Algorithms 1 and 2 .

Algorithm 1 Projected subgradient descent algorithm for u_l.

Input: $\boldsymbol{x}^{[1]}_l, \boldsymbol{x}^{[2]}_l, \cdots, \boldsymbol{x}^{[m]}_l, \boldsymbol{y}, \theta_w, C_{l1}, C_{l2}, \cdots, C_{lm}, d_l, \sigma_l, \{\theta_{1v}\}_{v=1}^m$;
Output: u_l;
Step 1: Calculate $\boldsymbol{Q}_l$ using $\boldsymbol{Q}_l = \boldsymbol{P}_l^{-1}$;
Step 2: Calculate $\boldsymbol{\gamma}'_{il}$ and $\boldsymbol{\gamma}''_{il}$ using $\boldsymbol{\alpha}'_l$ and $\boldsymbol{\alpha}''_l$;
Step 3: Calculate $\boldsymbol{G}_{iil}$;
Step 4: Set $t = 1$;
Step 5: Repeat
  For $i = 1, 2, \ldots, n$:
    $\tilde{y}_{il} = y_i - \frac{u_l}{|\boldsymbol{A}_l|}\sum_{v=1}^{m}\sum_{u=1}^{m}\theta_{lv}A_{wul}y_{i(l-1)} + \boldsymbol{G}_{iil}\boldsymbol{\gamma}'_{il} - u_l\boldsymbol{G}_{iil}\boldsymbol{\gamma}''_{il}$;
    $d_i = \mathbb{I}\{y_i\tilde{y}_{il}\}$;
    $u_l = u_l - \frac{1}{\sqrt{t}}d_i\left(\boldsymbol{G}_{iil}\boldsymbol{\gamma}''_{il} + \frac{1}{|\boldsymbol{A}_l|}\sum_{v=1}^{m}\sum_{u=1}^{m}\theta_{lv}A_{wul}y_{i(l-1)}\right)$;
    If $u_l > B$, **then** $u_l \leftarrow B$;
    End if
  $u_l \leftarrow \max(u_l, 0)$;
  $t \leftarrow t + 1$;
Step 6: Until convergence;
Output u_l;

4. Theoretical analysis

Within this section, we utilize Rademacher complexity theory (Bartlett & Mendelson, 2003) to verify the view-consistency and view-complementarity of MDCTM. Additionally, we examine the MDCTM's generalization error bound. In the end, we analyze the computational complexity of MDCTM. For a detailed definition of Rademacher complexity theory, please refer to Appendix A.

4.1. Consensus analysis

In the MDCTM, classifiers from different views are encouraged to reach consensus. Consider a sample set $S = \left\{\left(\boldsymbol{x}^{[1]}_i, \boldsymbol{x}^{[2]}_i, \cdots, \boldsymbol{x}^{[m]}_i, y_i\right)\right\}_{i=1}^n$ generated identically and independently (i.i.d.) according to a distribution D, define $\hat{\mathbb{E}} = \mathbb{E}_S$ and $\mathbb{E} = \mathbb{E}_D$ as the empirical expectation over S and the true expectation over D respectively. To theoretically assess this agreement, we define the consistency between any two predictors in the lth layer as:

$$\begin{aligned} f_{u,v} &= \left(\theta^*_{lu}\boldsymbol{w}^{*T}_{lu}\boldsymbol{\phi}_u\left(\boldsymbol{x}^{[u]}\right) - \theta^*_{lv}\boldsymbol{w}^{*T}_{lv}\boldsymbol{\phi}_v\left(\boldsymbol{x}^{[v]}\right)\right)^2 \\ &= \left(\sum_{i=1}^{n}\theta^*_{lu}\alpha^{[u]*}_{li}\boldsymbol{\kappa}_u(\boldsymbol{x}^{[u]}_{li}, \boldsymbol{x}^{[u]}) - \sum_{i=1}^{n}\theta^*_{lv}\alpha^{[v]*}_{li}\boldsymbol{\kappa}_v(\boldsymbol{x}^{[v]}_{li}, \boldsymbol{x}^{[v]})\right)^2. \end{aligned} \tag{45}$$

Theorem 1. *Let $\delta \in (0,1)$ and $M \in \mathbb{R}^+$. Consider an n-size m-view training set in lth layer, denoted as $S = \left\{\left(\boldsymbol{x}^{[1]}_i, \boldsymbol{x}^{[2]}_i, \cdots, \boldsymbol{x}^{[m]}_i, y_i\right)\right\}_{i=1}^n$ sampled independently from a distribution D. For any two distinct views, in other words, uth view and vth view, suppose the optimal $\boldsymbol{\alpha}^{[u]*}_l$, $\boldsymbol{\alpha}^{[v]*}_l$, θ^*_{lu}, and θ^*_{lv} derived from the MDCTM satisfy the following boundedness condition: $\left\|\theta^{*2}_{lu}\sum_{i,j=1}^{n}\alpha^{[u]*}_{li}\alpha^{[u]*}_{lj}\boldsymbol{\kappa}_p(\boldsymbol{x}^{[u]}_{li}, \boldsymbol{x}^{[u]}_{lj}) + \theta^{*2}_{lv}\sum_{i,j=1}^{n}\alpha^{[v]*}_{li}\alpha^{[v]*}_{lj}\boldsymbol{\kappa}_v(\boldsymbol{x}^{[v]}_{li}, \boldsymbol{x}^{[v]}_{lj})\right\|_F \le M$ and the kernel functions $\boldsymbol{\kappa}_u$ and $\boldsymbol{\kappa}_v$ are bounded in the feature spaces,*

Algorithm 2 Learning algorithm of MDCTM.

Input: $S = \{(\boldsymbol{x}_i^{[1]}, \boldsymbol{x}_i^{[2]}, \cdots, \boldsymbol{x}_i^{[m]}, y_i)\}_{i=1}^{i=n}$ and search interval for C_{lm}, d_l and σ;
Output: $\boldsymbol{F}_L = (f_L(\boldsymbol{x}_1), f_L(\boldsymbol{x}_2), \cdots, f_L(\boldsymbol{x}_n))^T, f_L(x_i) \in \{-1, +1\}$;
Procedure
Step 1: $l = 1$
1.1 Randomly choose the regularization parameters $C_{11}, C_{12}, \cdots, C_{1m}$, consistency parameter d_1, and kernel width
1.2 Give weight parameters $\{\theta_{1v}\}_{v=1}^m$ and $k_1 = 1$;
Repeat
1.3 Construct the optimization problem (3) and update $[\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2^*, \ldots, \boldsymbol{\alpha}_m^*]^T$;
1.4 Construct the optimization problem (8) and update $\{\boldsymbol{w}_{1v}\}_{v=1}^m$;
1.5 $k_1 \leftarrow k_1 + 1$;
Until $\{\theta_{1v}\}_{v=1}^m$ converged;
1.6 Obtain the predicted output $\boldsymbol{F}_1$;
Step 2: $l = 2, 3, \cdots, L$
2.1 $\boldsymbol{X}_l = \boldsymbol{X}_{l-1} \oplus \boldsymbol{F}_{l-1}$;
2.2 Randomly choose the regularization parameters $C_{l1}, C_{l2}, \cdots, C_{lm}$, consistency parameter d_l, and kernel width σ_l from the given interval ranges;
2.3 Give weight parameters $\{\theta_{lv}\}_{v=1}^m$ and $k_l = 1$, construct optimization problem (11);
Repeat
2.4 Apply Algorithm 1 on $\boldsymbol{X}_l$ and obtain u_l;
2.5 Update $[\boldsymbol{\alpha}_{11}^*, \boldsymbol{\alpha}_{l2}^*, \ldots, \boldsymbol{\alpha}_{lm}^*]^T$ using Eq. (18);
2.6 Construct optimization problem (23) and update $\{\boldsymbol{w}_{lv}\}_{v=1}^m$ according to Eq. (13);
2.7 $k_l \leftarrow k_l + 1$;
Until $\{\theta_{lv}\}_{v=1}^m$ converged;
2.8 Predicted output $\boldsymbol{F}_l$
Step 3: Calculate $\Delta_{F_l} = \|\boldsymbol{F}_l - \boldsymbol{F}_{l-1}\|_F^2$;
Step 4: If $\Delta_{F_l} \le \varepsilon$ (a given threshold);
End else
Step 5: $l \leftarrow l + 1$;
Output $\boldsymbol{F}_L$;

then with the probability $1-\delta$ *over* S*, the expected value of* $f_{u,v}$ *with* $\boldsymbol{\alpha}_l^{[u]*}$ *and* $\boldsymbol{\alpha}_l^{[v]*}$ *on new data is bounded by*

$$\mathbb{E}_D[f_{u,v}] \le \frac{1}{n}\sum_{i=1}^{n}\left(\xi_{li}^{[u]*} - \xi_{li}^{[v]*}\right)^2 + 3MR\sqrt{\frac{\ln(2/\delta)}{2n}} + \frac{4M}{n}\sqrt{\sum_{i=1}^{n}\left(\boldsymbol{\kappa}_u(\boldsymbol{x}_{li}^{[u]}, \boldsymbol{x}_{li}^{[u]}) + \boldsymbol{\kappa}_v(\boldsymbol{x}_{li}^{[v]}, \boldsymbol{x}_{li}^{[v]})\right)^2}, \tag{46}$$

where $\xi_{li}^{[u]*} = y_i - \sum_{j=1}^n \alpha_{lj}^{[u]*}\boldsymbol{\kappa}_u(\boldsymbol{x}_{lj}^{[u]}, \boldsymbol{x}_i^{[u]})$, $\xi_{li}^{[v]*} = y_i - \sum_{j=1}^n \alpha_{lj}^{[v]*}\boldsymbol{\kappa}_v(\boldsymbol{x}_{lj}^{[v]}, \boldsymbol{x}_i^{[v]})$*, and* $R = \max_{(\boldsymbol{x}^{[u]}, \boldsymbol{x}^{[v]}) \in \mathrm{supp}(D)} \left(\boldsymbol{\kappa}_v(\boldsymbol{x}^{[u]}, \boldsymbol{x}^{[u]}) + \boldsymbol{\kappa}_u(\boldsymbol{x}^{[v]}, \boldsymbol{x}^{[v]})\right)$.

Proof. Define the function class $\mathcal{F}_M$ as

$$\mathcal{F}_M = \left\{ f \mid f : (\boldsymbol{x}^{[u]}, \boldsymbol{x}^{[v]}) \to \left|\theta_{lu}\boldsymbol{w}_{lu}^\top\boldsymbol{\phi}_u(\boldsymbol{x}^{[u]}) - \theta_{lv}\boldsymbol{w}_{lv}^\top\boldsymbol{\phi}_v(\boldsymbol{x}^{[v]})\right|^2 = \left|\sum_{i=1}^{n}\theta_{lu}\alpha_{li}^{[u]}\boldsymbol{\kappa}_u(\boldsymbol{x}_{li}^{[u]}, \boldsymbol{x}^{[u]}) - \sum_{i=1}^{n}\theta_{lv}\alpha_{li}^{[v]}\boldsymbol{\kappa}_v(\boldsymbol{x}_{li}^{[v]}, \boldsymbol{x}^{[v]})\right|^2, \left\|\sum_{i,j=1}^{n}\left(\theta_{lu}^2\alpha_{li}^{[u]}\alpha_{lj}^{[u]}\boldsymbol{\kappa}_u(\boldsymbol{x}_{li}^{[u]}, \boldsymbol{x}_{lj}^{[u]}) + \theta_{lv}^2\alpha_{li}^{[v]}\alpha_{lj}^{[v]}\boldsymbol{\kappa}_v(\boldsymbol{x}_{li}^{[v]}, \boldsymbol{x}_{lj}^{[v]})\right)\right\| \le M \right\},$$

and $\hat{\mathcal{F}}_M = \left\{\hat{f} \mid \hat{f} : (\boldsymbol{x}^{[u]}, \boldsymbol{x}^{[v]}) \to \mathcal{A}(f),\ f \in \mathcal{F}_M\right\} \subseteq \mathcal{A}\circ\mathcal{F}_M$, where $\mathcal{A}$ is a Lipschitz function mapping the reals to the reals satisfying $\mathcal{A}(0) = 0$ defined as

$$\mathcal{A}(z) = \begin{cases} 0, & \text{if } z < 0; \\ \frac{z}{MR}, & \text{if } 0 \le z \le MR; \\ 1, & \text{otherwise,} \end{cases} \tag{47}$$

where $R = \max_{(\boldsymbol{x}^{[u]}, \boldsymbol{x}^{[v]}) \in \mathrm{supp}(D)} \left(\boldsymbol{\kappa}_u(\boldsymbol{x}_l^{[u]}, \boldsymbol{x}_l^{[u]}) + \boldsymbol{\kappa}_v(\boldsymbol{x}_l^{[v]}, \boldsymbol{x}_l^{[v]})\right)$. It is easy to get that the range of the functions in $\hat{\mathcal{F}}_M$ is $[0, 1]$.

Through some operations, the gap between the final predictors of each view $\left|\theta_{lu}^*\boldsymbol{w}_{lu}^{*\top}\boldsymbol{\phi}_u(\boldsymbol{x}_l^{[u]}) - \theta_{lv}^*\boldsymbol{w}_{lv}^{*\top}\boldsymbol{\phi}_v(\boldsymbol{x}_l^{[v]})\right|^2$ can be formulated as $\left(\hat{\boldsymbol{w}} \cdot \hat{\boldsymbol{\phi}}(\boldsymbol{x}_l)\right)$, where

$$\hat{\boldsymbol{w}} = \left[\mathrm{vec}(\theta_{lu}^{*2}\boldsymbol{w}_{lu}^*\boldsymbol{w}_{lu}^{*\top}),\ \mathrm{vec}(\theta_{lv}^{*2}\boldsymbol{w}_{lv}^*\boldsymbol{w}_{lv}^{*\top}),\ -\sqrt{2}\,\mathrm{vec}(\theta_{lu}^*\theta_{lv}^*\boldsymbol{w}_{lu}^*\boldsymbol{w}_{lv}^{*\top})\right]^\top, \tag{48}$$

$$\hat{\boldsymbol{\phi}}(\boldsymbol{x}) = \left[\mathrm{vec}(\boldsymbol{\phi}_u(\boldsymbol{x}^{[u]})\boldsymbol{\phi}_u(\boldsymbol{x}^{[u]})^\top),\ \mathrm{vec}(\boldsymbol{\phi}_v(\boldsymbol{x}^{[v]})\boldsymbol{\phi}_v(\boldsymbol{x}^{[v]})^\top),\ \sqrt{2}\,\mathrm{vec}(\boldsymbol{\phi}_u(\boldsymbol{x}^{[u]})\boldsymbol{\phi}_v(\boldsymbol{x}^{[v]})^\top)\right]^\top, \tag{49}$$

for convenience, we denote $\left(\boldsymbol{x}^{[u]}, \boldsymbol{x}^{[v]}\right)$ as $\boldsymbol{x}$, and $\mathrm{vec}(A)$ generates a row vector out of the entries of the matrix A by concatenating its rows. Define the Frobenius inner product $\circ$ between matrices $u_1u_2^\top \circ v_1v_2^\top = \left(\mathrm{vec}(u_1u_2^\top)\cdot\mathrm{vec}(v_1v_2^\top)\right) = v_1^\top u_1\, u_2^\top v_2$, then

$$\|\hat{\boldsymbol{w}}\|^2 = \left\|\theta_{lu}^{*2}\boldsymbol{w}_{lu}^*\boldsymbol{w}_{lu}^{*\top} + \theta_{lv}^{*2}\boldsymbol{w}_{lv}^*\boldsymbol{w}_{lv}^{*\top}\right\|_F^2 = \left\|\sum_{i,j=1}^{n}\left(\theta_{lu}^{*2}\alpha_{li}^{[u]*}\alpha_{lj}^{[u]*}\boldsymbol{\kappa}_u(\boldsymbol{x}_{li}^{[u]}, \boldsymbol{x}_{lj}^{[u]}) + \theta_{lv}^{*2}\alpha_{li}^{[v]*}\alpha_{lj}^{[v]*}\boldsymbol{\kappa}_v(\boldsymbol{x}_{li}^{[v]}, \boldsymbol{x}_{lj}^{[v]})\right)\right\|_F^2 \tag{50}$$

$$\hat{\boldsymbol{\kappa}}(\boldsymbol{x}, \boldsymbol{z}) = \hat{\boldsymbol{\phi}}(\boldsymbol{x})^\top\hat{\boldsymbol{\phi}}(\boldsymbol{z}) = \left(\boldsymbol{\kappa}_u(\boldsymbol{x}^{[u]}, \boldsymbol{z}^{[u]}) + \boldsymbol{\kappa}_v(\boldsymbol{x}^{[v]}, \boldsymbol{z}^{[v]})\right)^2. \tag{51}$$

With the optimal $\boldsymbol{\alpha}_l^{[u]*}$, $\boldsymbol{\alpha}_l^{[v]*}$, θ_{lu}^*, and θ_{lv}^* of MDCTM, $f_{u,v}$ varies in the range $[0, MR]$. Thus, $\hat{f}_{u,v} = \frac{f_{u,v}}{MR} = \mathcal{A}(f_{u,v}) \in \hat{\mathcal{F}}_M$ varies in $[0, 1]$ on the support of $\mathcal{D}$.

From Lemma 2 in Appendix A, the following inequality is satisfied with probability greater than $1-\delta$ over S,

$$\mathbb{E}_D[\hat{f}_{u,v}] \le \hat{\mathbb{E}}[\hat{f}_{u,v}] + \hat{R}_n(\hat{\mathcal{F}}_M) + 3\sqrt{\frac{\ln(2/\delta)}{2n}}. \tag{52}$$

Since $\mathcal{A}$ be a Lipschitz function with Lipschitz constant $1/(RM)$, the following inequality holds according to Lemmas 2 and 3 in Appendix A,

$$\hat{R}_n(\hat{\mathcal{F}}_M) \le \frac{4}{nR}\sqrt{\sum_{i=1}^{n}\left(\boldsymbol{\kappa}_u(\boldsymbol{x}_{li}^{[u]}, \boldsymbol{x}_{li}^{[u]}) + \boldsymbol{\kappa}_v(\boldsymbol{x}_{li}^{[v]}, \boldsymbol{x}_{li}^{[v]})\right)^2}. \tag{53}$$

We can obtain

$$\mathbb{E}_D[\hat{f}_{u,v}] \le \hat{\mathbb{E}}[\hat{f}_{u,v}] + 3\sqrt{\frac{\ln(2/\delta)}{2n}} + \frac{4}{nR}\sqrt{\sum_{i=1}^{n}\left(\boldsymbol{\kappa}_p(\boldsymbol{x}_{li}^{[p]}, \boldsymbol{x}_{li}^{[p]}) + \boldsymbol{\kappa}_q(\boldsymbol{x}_{li}^{[q]}, \boldsymbol{x}_{li}^{[q]})\right)^2}. \tag{54}$$

Due to $f_{u,v} = \hat{f}_{u,v} \cdot MR$, we have

$$\mathbb{E}_D[f_{u,v}] \le \hat{\mathbb{E}}[f_{u,v}] + 3MR\sqrt{\frac{\ln(2/\delta)}{2n}} + \frac{4M}{n}\sqrt{\sum_{i=1}^{n}\left(\boldsymbol{\kappa}_u(\boldsymbol{x}_{li}^{[u]}, \boldsymbol{x}_{li}^{[u]}) + \boldsymbol{\kappa}_v(\boldsymbol{x}_{li}^{[v]}, \boldsymbol{x}_{li}^{[v]})\right)^2}. \tag{55}$$

From the first constraint in Eq. (11) of the MDCTM with optimal $\boldsymbol{\alpha}_l^{[u]*}$, $\boldsymbol{\alpha}_l^{[v]*}$, we obtain

$$\hat{\mathbb{E}}[f_{u,v}] \le \frac{1}{n}\sum_{i=1}^{n}\left(\xi_{li}^{[u]*} - \xi_{li}^{[v]*}\right)^2, \tag{56}$$

where $\xi_{li}^{[u]*} = y_i - \sum_{j=1}^{n} \alpha_{lj}^{[u]*} \boldsymbol{\kappa}_u(\boldsymbol{x}_{lj}^{[u]}, \boldsymbol{x}^{[u]})$, $\xi_{li}^{[v]*} = y_i - \sum_{j=1}^{n} \alpha_{lj}^{[v]*} \boldsymbol{\kappa}_v(\boldsymbol{x}_{lj}^{[v]}, \boldsymbol{x}^{[v]})$.

By combining Eqs. (55) and (56), we arrive at the desired conclusion of the theorem. □

From the above theorem, the consensus principle in the MDCTM is ensured with a tight error bound when $n \to \infty$.

4.2. Complementarity analysis

In order to discuss the realization of complementarity in MDCTM, we also consider a sample set $S = \left\{ \left(\boldsymbol{x}_i^{[1]}, \boldsymbol{x}_i^{[2]}, \cdots, \boldsymbol{x}_i^{[m]}, y_i \right) \right\}_{i=1}^{n}$ generated identically and independently (i.i.d.) according to a distribution $\mathcal{D}$, define $\hat{\mathbb{E}} = \mathbb{E}_S$ and $\mathbb{E} = \mathbb{E}_{\mathcal{D}}$ as the empirical expectation over S and the true expectation over $\mathcal{D}$ respectively. In the first half of Section 5.5, our numerical experiments confirm that there exists a negative correlation between $\|\boldsymbol{w}_v\|^2$ and θ_v for v-th view. It can be interpreted as a smaller $\|\boldsymbol{w}_v\|^2$ indicates a larger margin and better generalization, leading to a higher reliability of the view and thus a larger weight θ_v. On this basis, we define the complementarity between any two hyperplanes in the lth layer as:

$$
\begin{aligned}
g_{u,v} &= \left(\theta_{lu}^* \left\| \boldsymbol{w}_{lu}^* \right\|^2 - \theta_{lv}^* \left\| \boldsymbol{w}_{lv}^* \right\|^2 \right)^2 \\
&= \left(\sum_{i,j=1}^{n} \left(\theta_{lu}^* \alpha_{li}^{*[u]} \alpha_{lj}^{*[u]} \boldsymbol{\kappa}_u(\boldsymbol{x}_{li}^{[u]}, \boldsymbol{x}_{lj}^{[u]}) - \theta_{lv}^* \alpha_{li}^{*[v]} \alpha_{lj}^{*[v]} \boldsymbol{\kappa}_v(\boldsymbol{x}_{li}^{[v]}, \boldsymbol{x}_{lj}^{[v]}) \right) \right)^2 .
\end{aligned} \tag{57}
$$

Theorem 2. *Let $\delta \in (0,1)$ and $M \in \mathbb{R}^+$. Consider an n-size m-view training set in lth layer, denoted as $S = \left\{ \left(\boldsymbol{x}_i^{[1]}, \boldsymbol{x}_i^{[2]}, \cdots, \boldsymbol{x}_i^{[m]}, y_i \right) \right\}_{i=1}^{n}$ sampled independently from a distribution $\mathcal{D}$. For any two distinct views, we denote as uth view and vth view, suppose the optimal $\boldsymbol{\alpha}_l^{[u]*}$, $\boldsymbol{\alpha}_l^{[v]*}$, θ_{lu}^*, and θ_{lv}^* derived from the MDCTM satisfy the following boundedness condition: $\left\| \sum_{i,j=1}^{n} (\theta_{lu}^{*2} \alpha_{li}^{[u]*} \alpha_{lj}^{[u]*} \boldsymbol{\kappa}_u(\boldsymbol{x}_{li}^{[u]}, \boldsymbol{x}_{lj}^{[u]}) + \theta_{lv}^{*2} \alpha_{li}^{[v]*} \alpha_{lj}^{[v]*} \boldsymbol{\kappa}_v(\boldsymbol{x}_{li}^{[v]}, \boldsymbol{x}_{lj}^{[v]})) \right\|_F \le M$ and the kernel functions $\boldsymbol{\kappa}_u$ and $\boldsymbol{\kappa}_v$ are bounded in the feature spaces, then with the probability $1-\delta$ over S, the expected value of $g_{u,v}$ with $\boldsymbol{\alpha}_l^{[u]*}$ and $\boldsymbol{\alpha}_l^{[v]*}$ is bounded by*

$$
\begin{aligned}
\mathbb{E}_{\mathcal{D}}[g_{u,v}] \le \frac{1}{n} \sum_{i=1}^{n} & \left(\frac{1}{\lambda_l} \left(\left(\frac{\lambda_l}{m} + \frac{B}{2} + u_l C \right) \Delta_W + u_l C B + \frac{B}{2} \Delta_W \right) \right)^2 \\
& + 3M\tau \sqrt{\frac{\ln(2/\delta)}{2n}} \\
& + \frac{4M}{n} \sqrt{\left(\boldsymbol{w}_{lu}^* \boldsymbol{w}_{lu}^{*\top} + \boldsymbol{w}_{lv}^* \boldsymbol{w}_{lv}^{*\top} \right)^2},
\end{aligned} \tag{58}
$$

where $B = \max_v \left\| \boldsymbol{w}_{lv}^* \right\|^2$, $C = \max_v \left| \sum_i c_{lv}^i \right|$, $\Delta_W = \max_{u,v} \left| \left\| \boldsymbol{w}_{lu}^* \right\|^2 - \left\| \boldsymbol{w}_{lv}^* \right\|^2 \right|$, *and* $\tau = \max_{(\boldsymbol{x}_l^{[u]}, \boldsymbol{x}_l^{[v]}) \in \mathrm{supp}(D)} \left(\sum_{i,j=1}^{n} \alpha_{li}^{[u]*} \alpha_{lj}^{[u]*} \boldsymbol{\kappa}_u(\boldsymbol{x}_{li}^{[u]}, \boldsymbol{x}_{lj}^{[u]}) + \sum_{i,j=1}^{n} \alpha_{li}^{[v]*} \alpha_{lj}^{[v]*} \boldsymbol{\kappa}_v(\boldsymbol{x}_{li}^{[v]}, \boldsymbol{x}_{lj}^{[v]}) \right)$.

Proof. Define the function class $\mathcal{G}_M$ as

$$
\begin{aligned}
\mathcal{G}_M = \Bigg\{ & g \mid g : (\boldsymbol{x}_l^{[u]}, \boldsymbol{x}_l^{[v]}) \to \left| \theta_{lu} \left\| \boldsymbol{w}_{lu} \right\|^2 - \theta_{lv} \left\| \boldsymbol{w}_{lv} \right\|^2 \right|^2 \\
& = \left| \sum_{i,j=1}^{n} \theta_{lu} \alpha_{li}^{[u]} \alpha_{lj}^{[u]} \boldsymbol{\kappa}_u(\boldsymbol{x}_{li}^{[u]}, \boldsymbol{x}_{lj}^{[u]}) - \sum_{i,j=1}^{n} \theta_{lv} \alpha_{li}^{[v]} \alpha_{lj}^{[v]} \boldsymbol{\kappa}_v(\boldsymbol{x}_{li}^{[v]}, \boldsymbol{x}_{lj}^{[v]}) \right|^2, \\
& \left\| \sum_{i,j=1}^{n} (\theta_{lu}^{*2} \alpha_{li}^{[u]*} \alpha_{lj}^{[u]*} \boldsymbol{\kappa}_u(\boldsymbol{x}_{li}^{[u]}, \boldsymbol{x}_{lj}^{[u]}) + \theta_{lv}^{*2} \alpha_{li}^{[v]*} \alpha_{lj}^{[v]*} \boldsymbol{\kappa}_v(\boldsymbol{x}_{li}^{[v]}, \boldsymbol{x}_{lj}^{[v]})) \right\|_F \le M \Bigg\},
\end{aligned}
$$

and $\widehat{\mathcal{G}}_M = \left\{ \hat{g} \mid \hat{g} : (\boldsymbol{x}_l^{[u]}, \boldsymbol{x}_l^{[v]}) \to \mathcal{B}(g),\ g \in \mathcal{G}_M \right\} \subseteq \mathcal{B} \circ \mathcal{G}_M$, where $\mathcal{B}$ is a Lipschitz function mapping the reals to the reals satisfying $\mathcal{B}(0) = 0$ defined as

$$
\mathcal{B}(x) = \begin{cases} 0, & \text{if } x < 0; \\ \frac{x}{\tau M}, & \text{if } 0 \le x \le M; \\ 1, & \text{otherwise.} \end{cases} \tag{59}
$$

Eq. (59) shows that the range of the functions in $\widehat{\mathcal{G}}_M$ is $[0,1]$.

The $\left(\theta_{lu} \left\| \boldsymbol{w}_{lu} \right\|^2 - \theta_{lv} \left\| \boldsymbol{w}_{lv} \right\|^2 \right)^2$ which we defined can be written as $\left(\hat{\boldsymbol{w}} \cdot \hat{\boldsymbol{\Phi}}(\boldsymbol{x}_l) \right)$, where $\hat{\boldsymbol{w}}$ is defined as in Eq. (48), and

$$
\hat{\boldsymbol{\Phi}}(\boldsymbol{x}_l) = \left[\mathrm{vec}(\boldsymbol{w}_{lu}^* \boldsymbol{w}_{lu}^{*\top}),\ \mathrm{vec}(\boldsymbol{w}_{lv}^* \boldsymbol{w}_{lv}^{*\top}),\ \sqrt{2}\, \mathrm{vec}(\boldsymbol{w}_{lu}^* \boldsymbol{w}_{lv}^{*\top}) \right]^\top, \tag{60}
$$

In addition, we get $\|\hat{\boldsymbol{w}}\|^2$ as Eq. (50), and

$$
\hat{\boldsymbol{\Omega}}(\boldsymbol{x}_l, \boldsymbol{z}_l) = \hat{\boldsymbol{\Phi}}(\boldsymbol{x}_l)^\top \hat{\boldsymbol{\Phi}}(\boldsymbol{z}_l) = \left(\boldsymbol{w}_{lu}^* \boldsymbol{w}_{lu}^{*\top} + \boldsymbol{w}_{lv}^* \boldsymbol{w}_{lv}^{*\top} \right)^2. \tag{61}
$$

With the optimal $\boldsymbol{\alpha}_l^{[u]*}$, $\boldsymbol{\alpha}_l^{[v]*}$, θ_{lu}^*, and θ_{lv}^* of MDCTM, $g_{u,v}$ varies in the range $[0, M\tau]$. Thus, $\hat{g}_{u,v} = \frac{g_{u,v}}{M\tau} = \mathcal{B}(g_{u,v}) \in \hat{\mathcal{G}}_M$ varies in $[0,1]$ on the support of $\mathcal{D}$.

According to Lemma 2 in Appendix A, the following inequality holds with probability at least $1-\delta$ over the sample S,

$$
\mathbb{E}_{\mathcal{D}}[\hat{g}_{u,v}] \le \hat{\mathbb{E}}[\hat{g}_{u,v}] + \hat{R}_n(\hat{\mathcal{G}}_M) + 3\sqrt{\frac{\ln(2/\delta)}{2n}}. \tag{62}
$$

Given that $\mathcal{B}$ is a Lipschitz-continuous function with constant $1/(\tau M)$, the following bound follows directly from Lemmas 2 and 3 in Appendix A,

$$
\hat{R}_n(\hat{\mathcal{G}}_M) \le \frac{4}{n\tau} \sqrt{\left(\boldsymbol{w}_{lu}^* \boldsymbol{w}_{lu}^{*\top} + \boldsymbol{w}_{lv}^* \boldsymbol{w}_{lv}^{*\top} \right)^2}. \tag{63}
$$

It follows that,

$$
\begin{aligned}
\mathbb{E}_{\mathcal{D}}[\hat{g}_{u,v}] \le & \hat{\mathbb{E}}[\hat{g}_{u,v}] + 3\sqrt{\frac{\ln(2/\delta)}{2n}} \\
& + \frac{4}{n\tau} \sqrt{\left(\boldsymbol{w}_{lu}^* \boldsymbol{w}_{lu}^{*\top} + \boldsymbol{w}_{lv}^* \boldsymbol{w}_{lv}^{*\top} \right)^2}.
\end{aligned} \tag{64}
$$

Due to $g_{u,v} = \hat{g}_{u,v} \cdot M\tau$, we have

$$
\begin{aligned}
\mathbb{E}_{\mathcal{D}}[g_{u,v}] \le & \hat{\mathbb{E}}[g_{u,v}] + 3M\tau \sqrt{\frac{\ln(2/\delta)}{2n}} \\
& + \frac{4M}{n} \sqrt{\left(\boldsymbol{w}_{lu}^* \boldsymbol{w}_{lu}^{*\top} + \boldsymbol{w}_{lv}^* \boldsymbol{w}_{lv}^{*\top} \right)^2}.
\end{aligned} \tag{65}
$$

From the KKT conditions for the optimization problem (23) of the MDCTM, we get

$$
\frac{1}{2} \left\| \boldsymbol{w}_{lv}^* \right\|^2 + \lambda_l \theta_{lv}^* - u_l \sum_i c_{lv}^i = \mu - \nu_v, \tag{66}
$$

where $c_{lv}^i = \left(\boldsymbol{w}_{lv}^T \boldsymbol{\phi}_v \left(\boldsymbol{x}_{li}^{[v]} \right) \right) y_{i(l-1)}$, $\nu_v \ge 0$ and μ are lagrange multipliers. Assume $\left\| \boldsymbol{w}_{lv}^* \right\|^2 \le B$, $\left| \sum_i c_{lv}^i \right| \le C$.

From the first constraint in optimization problem (23) of the MDCTM, we have $\mu = \frac{\lambda_l + \frac{1}{2} \sum_v \left\| \boldsymbol{w}_{lv}^* \right\|^2 - u_l \sum_v \sum_i c_{lv}^i}{m}$. Furthermore, we can deduce

$$
|\mu| \le \frac{\lambda_l + \frac{mB}{2} + m u_l C}{m}. \tag{67}
$$

Finally, we obtain

$$
\hat{\mathbb{E}}[g_{u,v}] \le \frac{1}{n} \sum_{i=1}^{n} \left(\frac{1}{\lambda_l} \left(\left(\frac{\lambda_l}{m} + \frac{B}{2} + u_l C \right) \Delta_W + u_l C B + \frac{B}{2} \Delta_W \right) \right)^2, \tag{68}
$$

where $\Delta_W = \max_{u,v} \left| \left\| \boldsymbol{w}_{lu}^* \right\|^2 - \left\| \boldsymbol{w}_{lv}^* \right\|^2 \right|$.

By combining Eqs. (65) and (68), we arrive at the conclusion of the theorem. □

Table 2
Details of all datasets.

Datasets	Ionosphere	Balance	100Leaves	Handwritten	AWA
View 1	Original view(34)	Original view(4)	Shape(64)	Fou(75)	PHOG(2000)
View 2	Extracted view(25)	Extracted view(4)	Edge(64)	Fac(215)	SIFT(252)
View 3	-	-	Texture(64)	Zer(46)	SURF(252)
Samples	351	625	1600	1200	2400
Classes	2	2	2	6	6

From the above theorem, the complementarity principle in the MDCTM is ensured with a tight error bound when $n \to \infty$.

4.3. Generalization error bound analysis

We can obtain the generalization error bound of MDCTM through the following theorem.

Theorem 3. *Given a constant* $\delta \in (0,1)$ *and* $I^{[1]}, I^{[2]}, \ldots, I^{[m]} \in \mathbb{R}^+$, *consider the training sample dataset* $S = \left\{ \left(\boldsymbol{x}_i^{[1]}, \boldsymbol{x}_i^{[2]}, \cdots, \boldsymbol{x}_i^{[m]}, y_i \right) \right\}_{i=1}^{i=n}$ *obtained by sampling independently and equally distributed from* $\mathcal{X} \times \mathcal{Y} = \boldsymbol{X}^{[1]} \times \boldsymbol{X}^{[2]} \times \cdots \times \boldsymbol{X}^{[m]} \times \boldsymbol{Y}$ *according to the probability distribution* $\mathcal{D}$, *define the decision function class in* l*th layer of MDCTM* $\mathcal{F}_l = \left\{ f_l \mid f_l : (\boldsymbol{x}_l^{[1]}, \boldsymbol{x}_l^{[2]}, \ldots, \boldsymbol{x}_l^{[m]}) \to \sum_{v=1}^{m} \theta_{lv} f_{lv}(\boldsymbol{x}_l^{[v]}) \right\}, \sum_{i,j=1}^{n} \alpha_{li}^{[v]} \alpha_{lj}^{[v]} \boldsymbol{\kappa}_v \left(\boldsymbol{x}_{li}^{[v]}, \boldsymbol{x}_{lj}^{[v]} \right) \le \left(I^{[v]} \right)^2$ *and* $\tilde{\mathcal{F}}_l = \left\{ \tilde{f}_l \mid \tilde{f}_l : (\boldsymbol{x}_l^{[1]}, \boldsymbol{x}_l^{[2]}, \ldots, \boldsymbol{x}_l^{[m]}, \boldsymbol{y}) \to -y f_l(\boldsymbol{x}_l^{[1]}, \boldsymbol{x}_l^{[2]}, \ldots, \boldsymbol{x}_l^{[m]}) \right\}$. *If the optimal solutions* $\left[\boldsymbol{\alpha}_{l1}^*, \boldsymbol{\alpha}_{l2}^*, \ldots, \boldsymbol{\alpha}_{lm}^* \right]^T$ *and* $\{\theta_{lv}^*\}_{v=1}^m$ *of MDCTM satisfy* $\sum_{i,j=1}^{n} \alpha_{li}^{[v]} \alpha_{lj}^{[v]} \boldsymbol{k}_v \left(\boldsymbol{x}_{li}^{[v]}, \boldsymbol{x}_{lj}^{[v]} \right) \le \left(I^{[v]} \right)^2$ *and the kernel function* $\{\boldsymbol{k}_v\}_{v=1}^m$ *is bounded, then the decision function (24) satisfies* $f_l(\boldsymbol{x}_l^{[1]}, \ldots, \boldsymbol{x}_l^{[m]}) \in \mathcal{F}_l$, *and the following holds with probability* $1-\delta$

$$P_D\left(\boldsymbol{y} f_l \le 0\right) \le \frac{1}{n} \sum_{i=1}^{n} \sum_{v=1}^{m} \theta_{lv}^* \left| \xi_{li}^{[v]} \right| + 3\sqrt{\frac{\ln(2/\delta)}{2n}} + \frac{4}{n} \sum_{v=1}^{m} \theta_{lv}^* I^{[v]} \sqrt{\sum_{i=1}^{n} \boldsymbol{k}_v \left(\boldsymbol{x}_{li}^{[v]}, \boldsymbol{x}_{li}^{[v]} \right)}, \tag{69}$$

where $\xi_{li}^{[v]} = y_i - \sum_{j=1}^{m} \alpha_{lj}^{[v]} \boldsymbol{k} \left(\boldsymbol{x}_{lj}^{[v]}, \boldsymbol{x}_{li}^{[v]} \right)$.

From the above theorem, we can conclude that when is δ fixed, $\sqrt{\frac{\ln(2/\delta)}{2n}}$ is a constant. The upper bound of the expected risk of MDCTM on the decision function (24) is related to the constant $I^{[v]}$, the number of views and the number of samples m. When the number of samples n is large enough, MDCTM has a tighter upper bound on the generalization error and a better generalization ability.

4.4. Computational complexity

One of the characteristics of MDCTM is its fast computational capability, since the fast leave-one-out cross-validation method is performed for parameter optimization in the above stacked architecture to determine the degree of cross-knowledge transfer between adjacent layers. Its complexity can be expressed as

$$T(n) = O((mn)^3) + O((L-1)((mn)^3 + n)) + O(km^2 L), \tag{70}$$

let m denote the number of views and n the number of samples. The initial term, $O\left((mn)^3\right)$, represents the computational complexity of constructing the MVLSSVM-2C model in the first layer, which involves solving the Lagrange multipliers in Eq. (18) through matrix inversion. The subsequent term, $O\left((L-1)\left((mn)^3 + n\right)\right)$, accounts for the computational complexity from the second layer onward, including the $O(n)$ complexity for each iteration of Algorithm 1. The third part, $O\left(km^2 L\right)$, is the time complexity of solving the convex quadratic optimization problem by the alternating optimization strategy from the first layer.

Let's use MV-LSSVM as an example to examine the computational complexity of traditional leave-one-out cross-validation. The theoretical computational complexity of MV-LSSVM is $O\left((mn)^3\right)$. When applying leave-one-out cross-validation, the complexity escalates to $O\left(n \cdot (mn)^3\right)$. Furthermore, performing a grid search over the regularization parameters C_1 (s_1 grid values), C_2 (s_2 grid values), D (s_3 grid values), and kernel width σ (s_4 grid values) increases the computational complexity of MV-LSSVM to $s_1 s_2 s_3 s_4 \cdot O\left(n \cdot (mn)^3\right)$. Typically, each of s_1, s_2, s_3, and s_4 exceeds three, whereas the number of layers L in MDCTM remains relatively small, ranging from $3 \le L \le 5$. Therefore, although the computational complexity of MDCTM may initially appear higher than that of MV-LSSVM, our experiments indicate that MV-LSSVM with grid search has a significantly longer running time compared to MDCTM.

Overall, the proposed MDCTM outperforms the traditional MV-LSSVM in terms of running speed.

5. Experiment

In this section, we evaluate the classification performance and running time of the proposed MDCTM in comparison to traditional SVM-based MVL models across multiple datasets. In addition, we performed a sensitivity analysis on the parameters of MDCTM. Furthermore, an ablation study was conducted to evaluate the individual contributions of the consistency term and the complementarity term. All experiments were conducted using PyCharm 2019.1.1 on a Windows 11 laptop equipped with an Intel Core i7 processor (2.90 GHz) and 16.00 GB of RAM.

5.1. Datasets

The details presented in Table 2 outline all datasets used in our experiment.

1) Ionosphere dataset and Balance Scale dataset[1]: Both datasets come from the UCI machine learning library. The ionosphere dataset comes from the Goose Bay radar system in Labrador, Canada. It classifies radar signals as "good" (indicating the presence of structure in the ionosphere) or "bad" (no structure). It contains 34 continuous features and 351 instances. The balance scale dataset is used to study the balance problem of the balance scale. It contains 4 features and 625 instances. Given that these two datasets are not multi-view, we refer to the original data as view A and to the 95 % principal components extracted from the original data as view B.

2) 100 Leaves dataset[1]: This dataset contains 1600 samples from 100 plants. Each sample has 64 features for describing leaves' shape, edge, and texture. To convert it into a binary classification task, we regard samples with label $1 \sim 10$ as positive samples and samples with label $11 \sim 20$ as negative samples. For the 3 views, 3 binary classification problems are generated.

3) Handwritten digit dataset[2]: Each class in the Handwritten digit dataset comprises 200 samples, totaling 2000 handwritten digit images ($0 \sim 9$). Every sample is characterized by six feature representations. For

[1] available at https://archive.ics.uci.edu/ml/index.php
[2] available at https://archive.ics.uci.edu/dataset/72/multiple+features



Fig. 3. AWA dataset.

our experiments, we select three pairs of digits: (1, 5), (2, 7), and (3, 8). We utilize 76-D Fourier coefficients of character shapes (fou), 216-D Profile correlations (fac), and 47-D Zernike moments (zer) to construct two views. This approach yields nine binary classifier problems.

4) AWA datasets[3]: This dataset consists of 6 pre-extracted feature representations, totaling 30,475 images of 50 animal categories. We selected 6 animal categories, namely chimpanzee, giant panda, humpback whale, leopard, rat, and seal (Fig. 3). And we select the first 200 samples in each animal category. For the 6 features in the dataset, we select 252-D Pyramid Histogram of Oriented Gradients (PHOG), 128-D Scale-Invariant Feature Transform (SIFT), and 2000-D L1 Normalized Speeded-Up Robust Features (SURF). We pair them into 45 binary datasets through a one-to-one strategy of perspetctive and animal category.

5.2. Comparison models

1) PSVM-2V (Tang et al., 2017): PSVM-2V is an enhanced variant derived from SVM-2K. It employs likeness constraints across various views to fulfill the consistency principle. Additionally, it utilizes distinct perspectives to furnish exclusive information to one another, thereby realizing the complementarity principle.

2) MV-LSSVM (Houthuys et al., 2018): MV-LSSVM is an extension of the LSSVM model on a multi-view dataset. It adds coupling terms to the original model to meet the complementarity principle and finally obtains the optimal hyperplane by solving a set of linear equations.

3) MVLSSVM-2C (Tang et al., 2022): MVLSSVM-2C minimizes the least squares loss of the training error and the prediction difference between different perspectives to meet the consistency principle. At the same time, it combines the weight distribution strategy to adaptively learn and adjust the importance of each perspective to fulfill the complementary principle.

4) MvSLMC (Zhao & Xu, 2024): MvSLMC incorporates view weights and structural information into the optimization problem, simultaneously satisfying the principles of consensus and complementarity.

5) MVLDM (Hu et al., 2024): MVLDM takes into account the margin distribution both within each view and across different views. It is implemented by leveraging the cross-view information of the multi-view margin distribution, specifically by maximizing the mean of the margin distribution between multiple views and minimizing the variance of the multi-view margins. Additionally, it upholds the consistency principle by enforcing a similarity constraint between the two views.

6) Mv$SL_{0/1}$-SVM (Chen et al., 2025): Mv$SL_{0/1}$-SVM introduces the ideal soft margin loss function for the first time in MVL and utilizes hierarchical agglomerative clustering to fuse structural information across views, following the principles of consistency and complementarity.

7) MvSTSVM (Liu et al., 2025): MvSTSVM introduces weights to multi-view datasets and uses clustering methods to mine intra-class and inter-class structural information on two hyperplanes, adhering to the principles of consistency and complementarity.

5.3. Experimental environment and model classification performance indicators

The experimental settings for all the MDCTM comparison methods are as follows: we use the grid search method to find the optimal parameters, and all algorithm kernel functions use Gaussian kernel functions. The search interval for all penalty parameters of PSVM-2V, MV-LSSVM, MVLSSVM-2C, MvSLMC, MVLDM, Mv$SL_{0/1}$-SVM, and MvSTSVM are set to $\{10^i \mid i = -2 : 1 : 2\}$. The parameter of the kernel function is selected from $\{10^i \mid i = -1 : 1 : 1\}$. In MvSLMC, Mv$SL_{0/1}$-SVM and MvSTSVM, the number of clusters is determined by the adaptive method. In particular, we set $\tau = 1.618$, $\varepsilon = 10^{-2}$, $l_m = 10$ and $K = 10^3$ in Mv$SL_{0/1}$-SVM, $C_+ = C_-$, $D_+ = D_-$, $\lambda = \eta$ in MvSTSVM. For the proposed MDCTM, the intervals of each parameter randomly selected are set to $C_{lv}, d_l \in \{10^i \mid i = -2 : 1 : 2\}$ and $\sigma \in \{10^i \mid i = -1 : 1 : 1\}$. Furthermore, for all experimental procedures, we employ a 5-fold cross-validation approach, which entails using 80 % of the data for training purposes and the remaining 20 % for testing.

In addition to the training time of each model, in order to more comprehensively evaluate the classification performance of the models, we use the *Accuracy*, *Precision*, *Recall*, and *F1_score* indicators. The following are the calculation formulas for these four indicators:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}, \tag{71}$$

$$Precision = \frac{TP}{TP + FP}, \tag{72}$$

$$Recall = \frac{TP}{TP + FN}, \tag{73}$$

$$F1_score = 2 \times \frac{Precision \times Recall}{Precision + Recall}. \tag{74}$$

In these formulas, TP represents the number of true positive samples, FN stands for the number of false negative samples, TN denotes the number of true negative samples, and FP refers to the number of false positive samples.

5.4. Experimental results

In this part of the discussion, we evaluate the performance of MDCTM against other SVM-based MVL models. After 5-fold cross-validation on 5 UCI datasets, 9 Handwritten datasets, and 45 AWA datasets, the mean and standard deviations of the models' accuracy and running time are shown in Tables 3-7. The results of other indicators are in Appendix B. The results verify the robustness of our proposed model on small and medium-sized datasets.

Fig. 4 displays the running times of all the methods compared using the AWA datasets, where the vertical axis represents the running time for each model and the horizontal axis denotes the index of the AWA datasets. In this context, the running time of MDCTM encompasses the

[3] available at https://cvml.ist.ac.at/AwA/

Table 3
Accuracy rates (%) with standard deviations for the UCI datasets.

Dataset	PSVM-2V	MV-LSSVM	MVLSSVM-2C	MvSLMC	MVLDM	Mv$SL_{0/1}$-SVM	MvSTSVM	MDCTM
Ionosphere	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
Balance	98.78 ± 1.17	83.30 ± 4.12	92.67 ± 5.64	**99.27 ± 3.43**	97.50 ± 1.25	98.87 ± 1.32	94.38 ± 3.24	99.13 ± 1.27
100Leaves_1	97.51 ± 2.90	94.66 ± 4.04	94.98 ± 4.25	97.42 ± 1.95	98.43 ± 0.09	**99.04 ± 2.65**	98.44 ± 2.12	98.50 ± 1.63
100Leaves_2	94.39 ± 3.72	94.04 ± 2.32	96.56 ± 1.52	98.00 ± 3.11	97.43 ± 0.77	96.91 ± 3.87	97.81 ± 1.44	**98.25 ± 1.43**
100Leaves_3	92.60 ± 4.73	95.92 ± 2.14	95.91 ± 1.92	95.99 ± 4.09	95.92 ± 2.14	96.43 ± 1.23	94.37 ± 7.45	**98.50 ± 0.56**
Avg. Acc.	95.82	91.98	95.03	97.67	97.32	97.81	96.25	**98.59**
Avg. Rank	6.00	7.13	6.50	3.00	4.58	2.75	4.75	**1.50**

Table 4
Accuracy rates (%) with standard deviations for the Handwritten datasets.

Dataset	PSVM-2V	MV-LSSVM	MVLSSVM-2C	MvSLMC	MVLDM	Mv$SL_{0/1}$-SVM	MvSTSVM	MDCTM
Handwritten_1	98.50 ± 3.37	97.13 ± 4.89	97.90 ± 3.89	99.21 ± 3.34	95.72 ± 2.72	99.37 ± 1.78	**100.00 ± 0.00**	99.49 ± 2.21
Handwritten_2	98.47 ± 2.46	97.97 ± 3.53	98.67 ± 2.22	96.89 ± 1.76	**100.00 ± 0.00**	99.93 ± 4.32	98.5 ± 3.23	**100.00 ± 0.00**
Handwritten_3	**98.64 ± 4.31**	96.72 ± 5.18	97.37 ± 4.59	97.92 ± 2.92	98.50 ± 2.24	98.01 ± 2.55	98.32 ± 1.34	98.49 ± 2.43
Handwritten_4	**100.00 ± 0.00**	96.34 ± 4.94	97.13 ± 4.89	**100.00 ± 0.00**	97.27 ± 2.72	98.54 ± 1.93	97.5 ± 4.32	**100.00 ± 0.00**
Handwritten_5	99.07 ± 1.96	94.80 ± 6.20	96.75 ± 6.01	98.01 ± 3.78	97.01 ± 1.93	97.91 ± 2.76	98.76 ± 2.22	**100.00 ± 0.00**
Handwritten_6	99.02 ± 2.06	98.94 ± 2.23	99.00 ± 3.16	99.91 ± 2.65	**100.00 ± 0.00**	**100.00 ± 0.00**	99.89 ± 1.76	99.49 ± 2.13
Handwritten_7	**100.00 ± 0.00**	98.43 ± 2.62	99.50 ± 1.58	99.95 ± 1.89	99.52 ± 1.51	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
Handwritten_8	**100.00 ± 0.00**	99.20 ± 1.58	**100.00** ± 0.00	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
Handwritten_9	98.67 ± 2.22	96.38 ± 5.25	96.62 ± 5.48	96.47 ± 2.55	98.05 ± 2.65	98.44 ± 1.21	96.47 ± 4.43	**98.99 ± 2.11**
Avg. Time	99.23	97.35	98.13	98.64	98.79	99.10	98.68	**99.62**
Avg. Rank	3.19	7.88	6.00	4.81	3.88	3.50	4.25	**2.50**

Table 5
Running time (s) for the UCI datasets.

Dataset	PSVM-2V	MV-LSSVM	MVLSSVM-2C	MvSLMC	MVLDM	Mv$SL_{0/1}$-SVM	MvSTSVM	MDCTM
Ionosphere	3398.7342	1196.0051	1223.1141	2433.3313	5219.7693	3942.3142	2220.3545	**245.3812**
Balance	11467.8346	3222.755	3808.0835	4092.3139	13453.2321	6542.4211	3345.4422	**769.2025**
100Leaves_1	13417.9845	3587.1508	3034.6647	3834.3231	10627.5432	4832.3139	3088.8741	**345.3421**
100Leaves_2	9971.5906	2845.2608	3096.4	3676.2131	12847.2157	5764.0948	3501.5323	**379.3322**
100Leaves_3	9780.5632	3627.8412	3225.5835	4032.9614	13218.8494	7646.0987	3564.8524	**312.3311**
Avg. Time	11159.4932	3320.7520	3291.1829	3908.9529	12536.7101	6196.2321	3375.1753	**451.5520**
Avg. Rank	7.25	3.00	2.75	5.00	7.75	6.00	3.25	**1.00**

Table 6
Running time (s) for the Handwritten datasets.

Dataset	PSVM-2V	MV-LSSVM	MVLSSVM-2C	MvSLMC	MVLDM	Mv$SL_{0/1}$-SVM	MvSTSVM	MDCTM
Handwritten_1	4650.4081	2402.3321	2693.9740	2312.3212	6832.2442	3232.3244	1122.8323	**702.2313**
Handwritten_2	2521.7162	1280.3921	1309.2932	2532.3144	4922.2939	2540.3244	1255.4353	**556.9821**
Handwritten_3	3535.3838	1648.2314	1687.2931	2640.1123	5493.3421	3072.5543	1176.8434	**565.2144**
Handwritten_4	3899.7041	2032.2341	2203.8724	2980.2234	5924.3414	2610.1123	1014.3233	**697.2414**
Handwritten_5	2938.5858	1493.2313	1503.3421	2870.5543	2042.4242	2324.3213	1332.4244	**656.2897**
Handwritten_6	1781.7999	593.4353	695.543	1632.4344	3845.3521	2054.4242	923.4344	**304.4355**
Handwritten_7	1546.8853	403.2441	470.3244	2424.3131	3032.9867	1943.2341	1043.3244	**359.2414**
Handwritten_8	2068.7599	389.2411	409.9833	1424.3221	2188.3242	1933.2313	1125.4340	**356.3243**
Handwritten_9	1384.4341	376.3243	387.3421	1032.1244	1942.4244	1487.4535	1231.1232	**198.3124**
Avg. Time	2459.6586	1027.0418	1083.3742	2192.0498	3673.9351	2245.7069	1137.7928	**463.7751**
Avg. Rank	6.37	2.50	3.50	5.63	7.63	6.25	3.12	**1.00**

total running time of each layer and the time needed to optimize the parameter u_l $(l = 2, 3, \ldots, L)$ in the lth layer. According to Fig. 4, MV-LSSVM and MVLSSVM-2C convert the classification task into a closed solution for solving a system of linear equations, so the running time is shorter. The MVLSSVM-2C adds an alternate optimization strategy for parameter iteration. PSVM-2V and MVLDM are solved by a large QPP, their running speeds are longer than that of MV-LSSVM and MVLSSVM-2C. However, MvSLMC, Mv$SL_{0/1}$-SVM and MvSTSVM need to apply hierarchical agglomerative clustering for sample clustering, which is slow to run under high-dimensional datasets such as AWA. The results verify the efficiency of our proposed model under high-dimensional datasets.

The capability to randomly assign parameters for each layer and efficiently determine the optimal value of u_l using a fast leave-one-out cross-validation method contributes significantly to the speed advantage demonstrated by these experimental results. Taking MV-LSSVM as an example, according to the analysis in 4.4, the computational complexity of MDCTM is given by $O\big((mn)^3\big) + O\big((L-1)\big((mn)^3 + n\big)\big) + O(km^2L)$. In contrast, the computational complexity of training MV-LSSVM with traditional cross-validation and grid search is $s_1 s_2 s_3 s_4 \cdot O\big(n \cdot (mn)^3\big)$. Therefore, the computational ratio of MV-LSSVM relative to the proposed methods is approximately $(s_1 s_2 s_3 s_4 / L)O(n)$, this is the reason why the running time of MV-LSSVM far exceeds that of the suggested approach.

Fig. 5 shows the changes in accuracy, precision, recall, and F1_score as the number of layers changes in the first 4 AWA datasets. It can be seen that for these 4 datasets, the second and third layers have a significant effect on improving the four indicators, the fourth layer has a

Table 7
Accuracy rates (%) with standard deviations for the AWA datasets.

Dataset	PSVM-2V	MV-LSSVM	MVLSSVM-2C	MvSLMC	MVLDM	Mv$SL_{0/1}$-SVM	MvSTSVM	MDCTM
AWA_1	85.25 ± 3.39	88.37 ± 6.42	89.75 ± 6.98	92.00±6.52	92.25 ± 1.63	91.32±5.43	93.23±2.45	**95.49 ± 3.49**
AWA_2	82.22 ± 5.46	89.47 ± 2.63	89.47 ± 2.63	93.25±2.59	93.25 ± 1.68	94.45±3.53	95.5±1.34	**98.99 ± 0.38**
AWA_2	82.22 ± 5.46	89.47 ± 2.63	89.47 ± 2.63	93.25±2.59	93.25 ± 1.68	94.45±3.53	95.50±1.34	**98.99 ± 0.38**
AWA_3	87.40 ± 4.63	63.39 ± 6.10	76.28 ± 7.98	91.25±2.93	85.50 ± 7.74	92.32±4.43	87.5±6.44	**94.04 ± 1.56**
AWA_4	83.68 ± 5.50	54.21 ± 12.92	67.02 ± 6.19	**91.75±1.07**	90.00 ± 3.85	85.32±1.98	82.50±3.65	88.47 ± 4.32
AWA_5	83.38 ± 5.85	88.72 ± 3.20	88.72 ± 3.20	95.75±2.09	93.50 ± 1.63	90.32±2.09	89.00±1.11	**98.49 ± 2.54**
AWA_6	88.74 ± 7.12	87.50 ± 2.34	82.25 ± 8.26	94.50±3.01	**97.75 ± 1.63**	94.22±8.21	88.5±5.45	95.48 ± 2.45
AWA_7	91.32 ± 3.54	89.74 ± 2.44	90.89 ± 3.51	89.75±2.05	95.25 ± 2.40	94.21±1.32	96.25±2.34	**98.99 ± 0.78**
AWA_8	82.10 ± 5.68	89.50 ± 3.71	90.75 ± 4.73	92.75±5.11	89.75 ± 3.11	**95.32±4.21**	93.32±4.21	94.48 ± 3.56
AWA_9	82.94 ± 5.25	88.47 ± 2.44	86.74 ± 4.30	93.75±1.25	94.50 ± 2.27	91.45±3.11	**94.54±1.33**	88.47 ± 2.44
AWA_10	90.77 ± 2.63	88.72 ± 4.23	90.66 ± 3.60	92.00±3.38	89.50 ± 3.26	91.32±2.99	89.75±3.31	**98.49 ± 1.35**
AWA_11	94.68 ± 2.67	90.75 ± 4.73	90.25 ± 1.05	88.25±3.38	85.75 ± 4.64	97.42±1.21	95.00±2.77	**99.24 ± 0.32**
AWA_12	84.53 ± 5.84	88.47 ± 4.34	89.75 ± 3.11	**93.75±6.84**	90.75 ± 4.73	88.32±5.32	91.75±4.54	92.98 ± 0.33
AWA_13	86.06 ± 3.37	88.97 ± 5.23	88.97 ± 5.23	88.5±1.05	92.50 ± 3.42	85.65±6.98	87.75±2.56	**96.74 ± 1.42**
AWA_14	88.27 ± 4.10	88.72 ± 4.08	88.72 ± 4.08	91.5±5.18	93.75 ± 3.85	96.99±0.21	94.56±1.22	**100.00 ± 0.00**
AWA_15	91.45 ± 3.53	89.72 ± 3.32	90.00 ± 3.75	93.25±2.88	94.25 ± 2.09	93.34±2.29	98.25±4.34	**100.00 ± 0.00**
AWA_16	90.64 ± 4.91	88.74 ± 2.45	88.96 ± 3.77	89.75±6.15	90.50 ± 3.71	93.02±3.45	**93.45±5.45**	92.98 ± 3.67
AWA_17	84.92 ± 6.61	88.97 ± 6.83	84.26 ± 4.07	94.25±5.42	89.26 ± 3.00	89.43±2.87	87.25±3.44	**96.98 ± 4.23**
AWA_18	91.22 ± 3.00	91.76 ± 4.58	92.62 ± 5.02	94.75±6.27	90.24 ± 2.83	89.43±4.22	92.43±1.98	**100.00 ± 0.00**
AWA_19	95.92 ± 3.61	89.21 ± 3.20	90.41 ± 2.29	93.59±2.41	93.25 ± 4.81	96.43±1.43	95.25±2.45	**97.75 ± 1.37**
AWA_20	91.78 ± 4.78	88.34 ± 3.50	92.12 ± 4.73	91.5±1.63	93.71 ± 4.44	95.25±2.45	93.75±3.99	**97.50 ± 1.77**
AWA_21	89.98 ± 4.97	88.46 ± 2.46	89.41 ± 3.56	**98.75±8.83**	92.32 ± 4.16	96.43±0.94	95.32±3.51	95.92 ± 3.61
AWA_22	87.43 ± 5.07	97.49 ± 1.34	89.47 ± 3.42	95.5±6.85	94.35 ± 3.65	95.04±2.12	97.25±5.19	**99.74 ± 0.12**
AWA_23	94.89 ± 3.39	91.44 ± 4.67	91.44 ± 4.67	96.00±1.63	92.34 ± 4.66	96.76 ± 2.31	93.42 ± 1.87	**98.50 ± 0.56**
AWA_24	91.67 ± 4.58	88.88 ± 4.05	92.34 ± 4.66	**94.80±3.71**	90.41 ± 4.29	93.21 ± 1.65	92.42 ± 2.14	94.75 ± 3.11
AWA_25	97.00 ± 1.12	86.38 ± 7.37	90.75 ± 7.53	91.25±4.43	94.75 ± 4.28	91.03 ± 2.89	90.21 ± 1.52	**98.25 ± 1.43**
AWA_26	91.00 ± 4.63	88.00 ± 8.78	88.00 ± 8.78	90.00±3.31	90.00 ± 6.79	96.12 ± 0.97	92.31 ± 2.63	**100.00 ± 0.00**
AWA_27	93.50 ± 2.71	85.40 ± 6.57	86.38 ± 7.37	95.25±1.05	92.50 ± 6.56	**100.00± 0.00**	96.21± 0.50	95.50 ± 2.44
AWA_28	84.23 ± 8.07	77.35 ± 5.99	88.71 ± 5.04	86.75±5.93	96.25 ± 1.53	87.43± 3.25	90.32± 2.78	**96.50 ± 2.40**
AWA_29	95.89 ± 4.47	89.93 ± 5.02	84.81 ± 6.31	93.25±6.93	92.25 ± 3.89	93.23 ± 1.77	93.5± 1.95	**97.24 ± 3.12**
AWA_30	95.00 ± 5.93	87.92 ± 4.18	88.01 ± 7.45	88.25±3.26	93.32 ± 5.64	93.32± 2.41	95.32± 0.89	**96.00 ± 3.58**
AWA_31	90.00 ± 6.12	87.04 ± 4.95	88.77 ± 5.69	91.50±5.03	91.98 ± 5.19	92.32± 3.12	**94.43± 1.67**	93.32 ± 5.64
AWA_32	97.75 ± 2.24	88.27 ± 6.28	89.39 ± 5.02	95.75±4.72	93.09 ± 4.29	96.43± 1.08	93.32± 2.32	**100.00 ± 0.00**
AWA_33	95.75 ± 4.01	90.25 ± 2.71	92.75 ± 2.71	90.75±3.60	96.75 ± 1.43	93.32± 2.75	91.21± 3.15	**97.02 ± 2.25**
AWA_34	83.75 ± 6.92	88.78 ± 5.54	90.00 ± 6.12	**92.00±4.21**	90.25 ± 4.79	86.21± 3.42	88.32± 2.86	90.72 ± 4.23
AWA_35	89.39 ± 5.02	89.01 ± 6.08	91.25 ± 3.95	89.75±3.59	96.75 ± 1.43	90.76± 1.93	90.32± 1.44	**98.00 ± 0.68**
AWA_36	89.44 ± 3.29	82.98 ± 12.79	82.98 ± 12.79	88.50±3.89	88.27 ± 6.28	87.43± 2.67	88.54± 3.21	**90.72 ± 4.21**
AWA_37	**95.75 ± 3.54**	88.87 ± 4.97	89.63 ± 4.92	90.71±6.65	94.92 ± 3.89	88.11± 1.58	89.23± 2.09	90.72 ± 1.67
AWA_38	86.40 ± 6.81	88.75 ± 1.98	87.94 ± 4.87	90.50±5.63	88.54 ± 5.98	90.53± 2.34	90.69± 1.77	**90.72 ± 2.45**
AWA_39	83.80 ± 6.55	87.11 ± 5.04	83.20 ± 5.25	85.25±5.47	88.42 ± 5.32	87.32± 3.05	**92.32± 0.92**	90.72 ± 8.34
AWA_40	92.75 ± 3.89	81.63 ± 6.10	79.82 ± 7.37	90.00±3.06	93.75 ± 2.93	91.31± 1.82	92.13± 2.45	**95.49 ± 2.06**
AWA_41	90.00 ± 6.79	82.10 ± 5.68	86.61 ± 4.59	87.59±7.09	86.00 ± 6.58	**95.32± 0.75**	90.32± 3.32	89.22 ± 6.45
AWA_42	88.75 ± 1.77	84.83 ± 3.77	81.51 ± 6.51	86.25±7.23	90.25 ± 2.71	85.21± 2.98	84.32± 3.45	**90.72 ± 4.24**
AWA_43	88.87 ± 4.94	80.55 ± 7.16	85.31 ± 3.65	90.27±10.73	89.39 ± 5.02	**94.87± 1.26**	92.31± 2.17	90.72 ± 6.33
AWA_44	89.89 ± 4.75	84.54 ± 4.01	85.36 ± 3.68	93.25±2.88	**94.00 ± 3.24**	90.32± 2.63	88.32± 1.88	89.22 ± 8.23
AWA_45	81.48 ± 8.94	84.07 ± 4.18	81.33 ± 3.47	90.25±6.89	90.54± 1.47	89.21± 2.71	89.75 ± 5.26	**90.72 ± 3.34**
Avg. Time	89.46	86.44	87.32	91.72	91.89	92.08	91.72	**95.24**
Avg. Rank	5.25	6.91	6.38	3.98	4.22	3.78	3.79	**1.69**

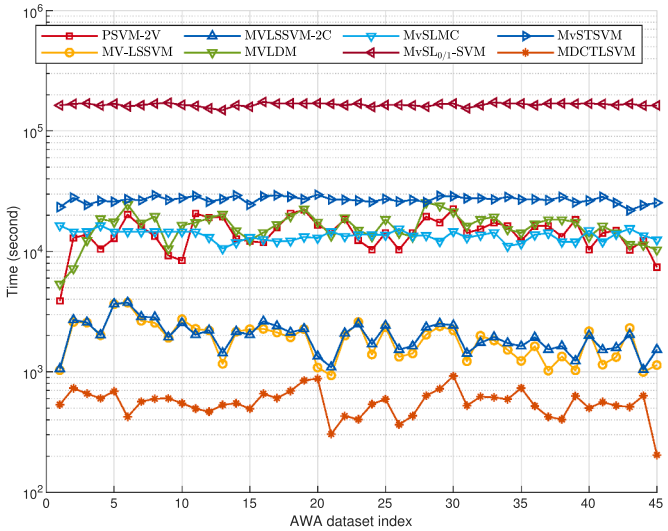


Fig. 4. Running time of 8 algorithms on AWA datasets.

more minor change than the third layer, and there is a downward trend starting from the fifth layer.

5.5. Parameter analysis

In order to explore the specific implementation of complementarity in MDCTM, we find the normal vector w_v^* and perspective weight θ_v^* of each view hyperplane corresponding to the optimal parameters in the last layer for the first 5 data sets in UCI and the first 3 data sets in AWA. As shown in Table 8, in these 8 data sets, $\|\boldsymbol{w}_v\|^2$ and θ_v are negatively correlated for the view v. This is because $\|\boldsymbol{w}_v\|^2$ determines the margin size of the classification hyperplane associated with view v. A smaller value of $\|\boldsymbol{w}_v\|^2$ implies a larger margin, indicating a smoother decision boundary and improved generalization capability. Consequently, the predictive output from this view is considered more reliable, and the corresponding weight θ_v is accordingly increased.

In addition, we explore the role of deep stacking architecture and transfer learning in improving the generalization ability of MDCTM, for the first 4 datasets in AWA, after selecting the optimal $\{\theta_{lv}\}_{v=1}^{m}$ through the alternating optimization strategy, we choose to split the search interval into smaller for parameters C_{1v}, C_{2v} and $d_l \in \{10^i \mid i = -3 : 1 : 3\}$

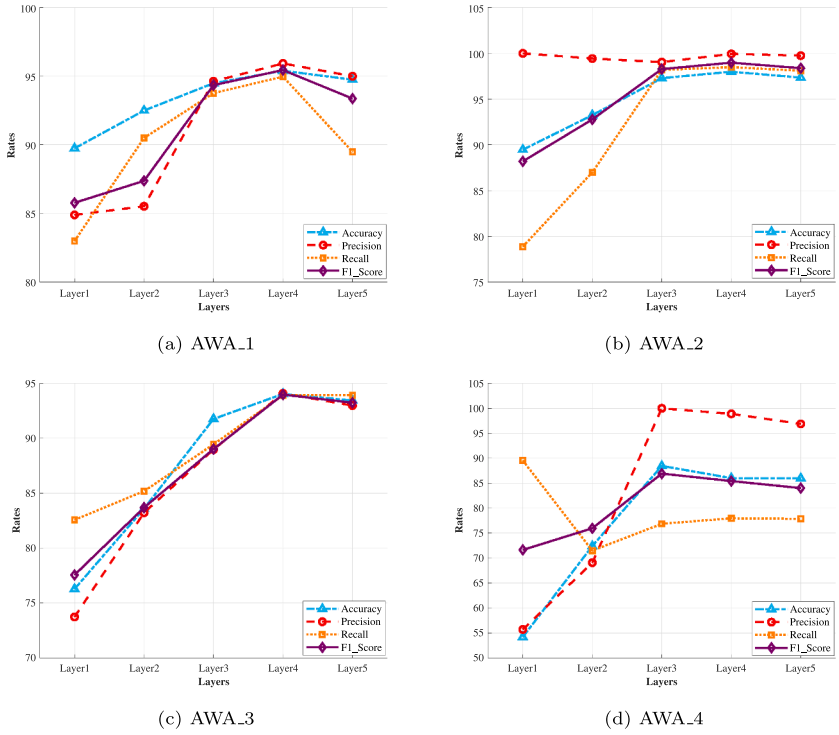


Fig. 5. Accuracies, precision, recall, and F1_score under change with the number of layers on the first 4 AWA datasets.

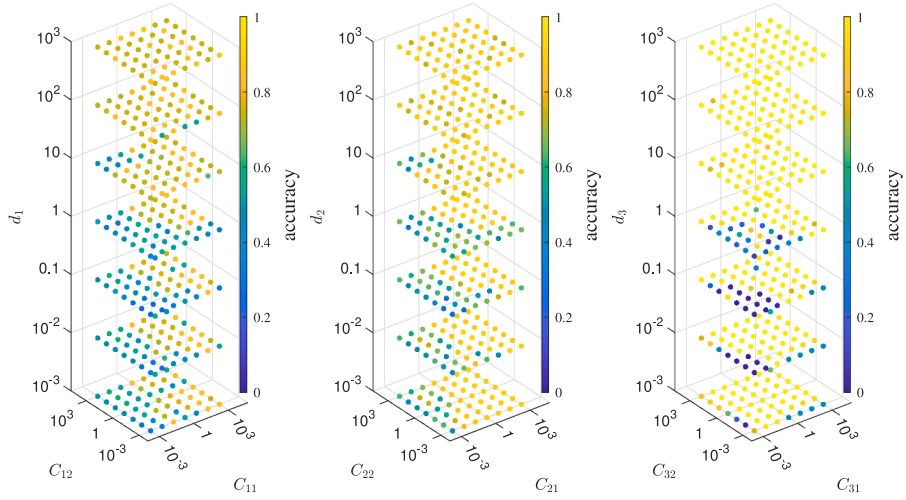


Fig. 6. The change of accuracy with the number of layers under $\sigma = 1$ on the AWA_1 dataset.

Table 8
Comparison of optimal $\|\boldsymbol{w}_v^*\|^2$ and θ_v^* in the last layer.

Dataset	Ionosphere	Balance	100Leaves_1	100Leaves_2	100Leaves_3	AWA_1	AWA_2	AWA_3
$\|\boldsymbol{w}_1^*\|^2$	1.4108	5.0203	35.615	34.3241	3.0221	0.0034	0.0048	1.7181
$\|\boldsymbol{w}_2^*\|^2$	1.1857	25.1953	47.57465	90.3477	33.6721	0.0021	0.0023	2.4849
θ_1^*	0.4968	0.6010	0.5001	0.6797	0.7233	0.4683	0.3092	0.5171
θ_2^*	0.5032	0.3990	0.4999	0.3203	0.2767	0.5317	0.6908	0.4829

under $\sigma = 1$, and obtain the accuracies of MDCTM under each parameter combination in the first 3 layers, as shown in Figs. 6-9, the color of each dot represents the accuracy corresponding to its parameter combination. We can intuitively see that as the number of layers increases, the improvement in accuracy for any parameter combination is almost significant, the accuracy corresponding to most of the parameter combinations in the third layer is close to 100 %, proving our model's robustness and reflecting the feasibility of our random selection of parameters rather than network search.

5.6. Ablation study

In order to verify the influence of consistency term and complementarity term in MDCTM, we compare the performance of MDCTM in

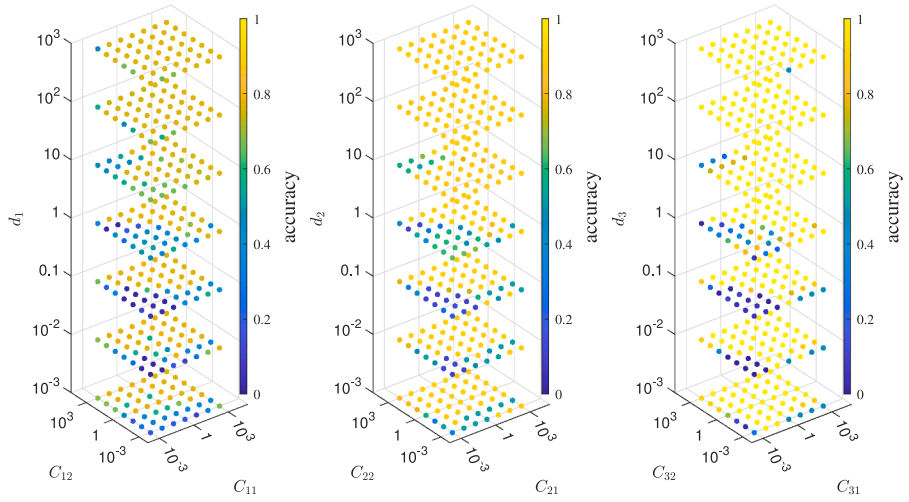


Fig. 7. The change of accuracy with the number of layers under $\sigma = 1$ on the AWA_2 dataset.

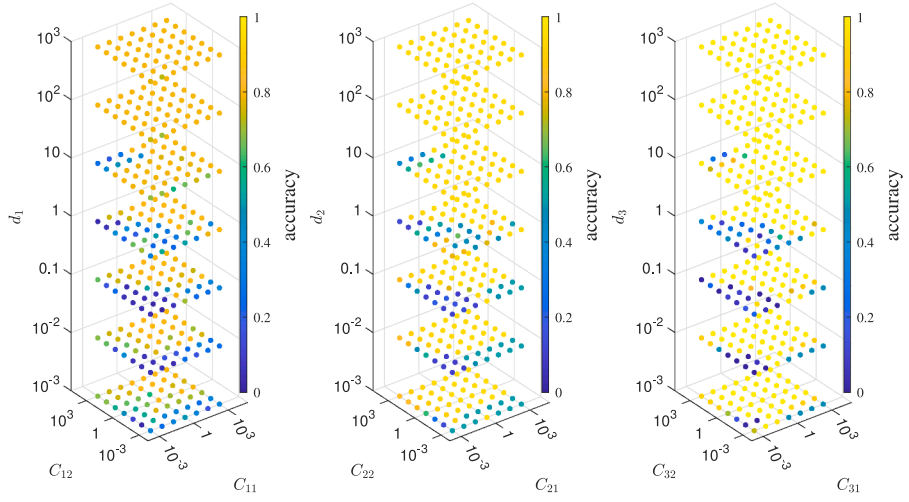


Fig. 8. The change of accuracy with the number of layers under $\sigma = 1$ on the AWA_3 dataset.

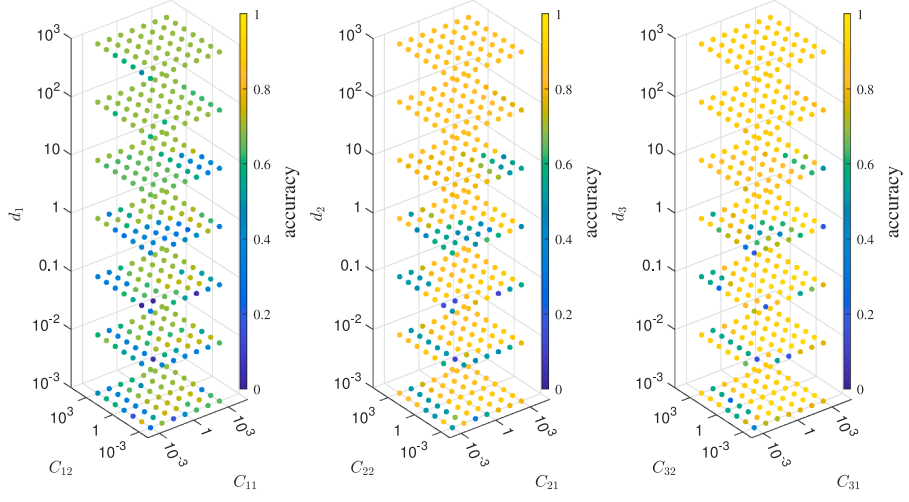


Fig. 9. The change of accuracy with the number of layers under $\sigma = 1$ on the AWA_4 dataset.

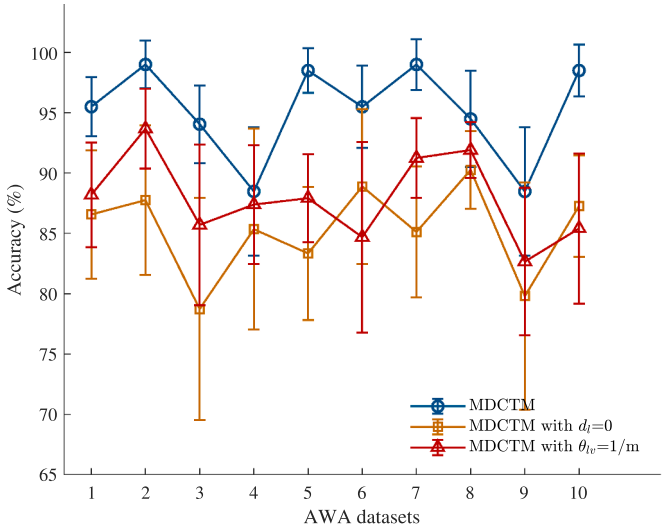


Fig. 10. Ablation experiments on MDCTM.

different cases on the first 10 AWA datasets: (1) MDCTM as in problem (10); (2) MDCTM with the parameters $d_l = 0$, that is, the model that only satisfies the principle of complementarity. (3) MDCTM with the parameters $\theta_{lv} = 1/2$, that is, the model that only satisfies the principle of consistency. In this experiment, except for the fixed parameters of MDCTM, the rest of the parameters adopt the optimal parameters to train and test the corresponding model.

The experimental results are shown in Fig. 10, where the horizontal axis is the dataset index and the vertical axis is the accuracy of the model in the corresponding case on the dataset. It can be clearly seen that the MDCTM in the first case performs best on all datasets. In addition, the MDCTM in the third case is usually more competitive than the MDCTM in the second case. This shows that the consistency principle is more instructive in MDCTM, and satisfying both principles at the same time can further improve the learning effect.

6. Conclusion and future work

This paper mainly proposes the MDCTM, which applies the deep cross-output knowledge transfer method based on the MV-LSSVM. It follows the principles of consistency and complementarity in MVL. Firstly, we use the MVLSSVM-2C as the first layer of the model. Starting from the second layer, we add the transfer learning influencing parameter and attach the predicted values of the previous layers to the sample attribute values. Then, we propose the optimization problem of the subsequent layers. To solve this problem, we propose the idea of fast leave-one-out cross-validation method to determine how much output knowledge is transferred between layers and randomly select model parameters such as regularization parameters, which effectively avoids the complex process of parameter network search and improves the solution speed. More importantly, the model incorporates deep stacking architecture and transfer learning, which can better handle complex problems to enhance the adaptability in new tasks and the classification performance of the model. Many numerical experimental results show that the proposed MDCTM improves SVM-based MVL models' running time and classification performance.

In this study, we discovered an interesting phenomenon, as shown in Figs. 6-9. The accuracy of MDCTM corresponding to most parameter combinations on the third layer has been significantly improved. However, a few parameter combinations have extremely low accuracy on the third layer. This phenomenon may increase the number of iteration layers while randomly selecting parameters, thereby prolonging the running time. We plan to explore this issue in our future work.

CRediT authorship contribution statement

Shuangrui Jia: Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization; **Sijie Liang:** Software, Validation, Writing – review & editing; **Ziyi Mo:** Formal analysis, Writing – review & editing; **Chunxiao Liu:** Writing – review & editing; **Huiru Wang:** Conceptualization, Writing – review & editing, Supervision, Project administration, Funding acquisition; **Chen Chen:** Validation, Writing – review & editing.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Rademacher complexity theory

Definition 1. Let $S = \{\boldsymbol{x}_1, \cdots, \boldsymbol{x}_n\}$ be a sample group drawn from a distribution D over the set $\mathcal{X}$, and let $\mathcal{F}$ denote a class of real-valued functions with domain $\mathcal{X}$. The empirical Rademacher complexity of $\mathcal{F}$ is then defined as the random variable

$$\hat{R}_n(\mathcal{F}) = \mathbb{E}_\sigma\left[\sup_{f\in\mathcal{F}}\frac{2}{n}\left|\sum_{i=1}^{n}\sigma_i f(\boldsymbol{x}_i)\right|\middle|\boldsymbol{x}_1,\ldots,\boldsymbol{x}_n\right], \tag{A.1}$$

where $\sigma = \{\sigma_1, \ldots, \sigma_n\}$ consists of independent Rademacher random variables uniformly taking values in $\{\pm 1\}$. The Rademacher complexity of $\mathcal{F}$ is

$$R_n(\mathcal{F}) = \mathbb{E}_S\left[\hat{R}_n(\mathcal{F})\right] = \mathbb{E}_{S\sigma}\left[\sup_{f\in\mathcal{F}}\left|\frac{2}{n}\sum_{i=1}^{n}\sigma_i f(\boldsymbol{x}_i)\right|\right]. \tag{A.2}$$

Lemma 1. *Let $\delta \in (0,1)$ and let $\mathcal{F}$ be a set of functions that map from the input space $\mathcal{X}$ to the interval [0,1]. Assume that $\{x_i\}_{i=1}^n$ are independently sampled from a probability distribution D. Then, with probability at least $1-\delta$ over the random selection of n samples, every function $f \in \mathcal{F}$ satisfies*

$$\mathbb{E}_D[f(\boldsymbol{x})] \le \hat{\mathbb{E}}_S[f(x)] + \hat{R}_n(\mathcal{F}) + 3\sqrt{\frac{\ln(2/\delta)}{2n}}. \tag{A.3}$$

Lemma 2. *Let $k : \mathcal{X}\times\mathcal{X} \to \mathbb{R}$ be a kernel, and let $S = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n\}$ be a sample group drawn from $\mathcal{X}$. The function class $\mathcal{F}_B = \{\boldsymbol{x} \to (\boldsymbol{w}\cdot\boldsymbol{\phi}(\boldsymbol{x})) : \|\boldsymbol{w}\| \le B\}$ is considered, where $\boldsymbol{\phi}$ is the feature mapping associated with the kernel k. The empirical Rademacher complexity of $\mathcal{F}_B$ meets the following criteria within the kernel feature space*

$$\hat{R}_n(\mathcal{F}_B) = \frac{2B}{n}\sqrt{\sum_{i=1}^{n}\boldsymbol{k}(\boldsymbol{x}_i,\boldsymbol{x}_i)}. \tag{A.4}$$

Lemma 3. *A Lipschitz function $\mathcal{A}$ with Lipschitz constant L, mapping the real numbers to real numbers and satisfying $\mathcal{A}(0) = 0$, is considered. The Rademacher complexity of the class $\mathcal{A}\circ\mathcal{F}$ satisfies*

$$\hat{R}_n(\mathcal{A}\circ\mathcal{F}) \le 2L\hat{R}_n(\mathcal{F}). \tag{A.5}$$

Appendix B. The results on precision, recall, and F1_score of five algorithms on numerous datasets

After 5-fold cross-validation on 5 UCI datasets, 9 Handwritten datasets, and 45 AWA datasets, the mean and standard deviations of the algorithms's precision, recall, and F1_score are shown in Tables B.9- B.17 respectively.

Table B.9
Precision rates (%) with standard deviations for the UCI datasets.

Dataset	PSVM-2V	MV-LSSVM	MVLSSVM-2C	MvSLMC	MVLDM	Mv$SL_{0/1}$-SVM	MvSTSVM	MDCTM
Ionosphere	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
Balance	98.61 ± 1.83	87.62 ± 6.73	93.08 ± 5.17	98.85 ± 3.12	99.09 ± 1.15	98.52 ± 1.25	93.52 ± 2.87	**99.12 ± 1.13**
100Leaves_1	95.38 ± 5.41	94.81 ± 3.97	95.01 ± 4.34	96.80 ± 1.82	98.52 ± 0.09	98.61 ± 2.48	97.89 ± 1.95	**98.99 ± 1.39**
100Leaves_2	89.75 ± 7.21	94.14 ± 2.23	96.46 ± 1.48	97.35 ± 2.95	97.06 ± 0.08	96.12 ± 3.65	96.95 ± 1.32	**98.50 ± 1.37**
100Leaves_3	89.36 ± 6.78	96.06 ± 2.13	96.07 ± 1.95	94.92 ± 3.88	96.06 ± 2.13	95.89 ± 1.15	92.14 ± 6.80	**100.00 ± 0.00**
Avg. Acc.	93.28	93.16	95.16	96.98	97.68	97.29	95.13	**99.15**
Avg. Rank	6.50	6.63	5.25	4.00	2.88	4.50	5.25	**1.00**

Table B.10
Recall rates (%) with standard deviations for the UCI datasets.

Dataset	PSVM-2V	MV-LSSVM	MVLSSVM-2C	MvSLMC	MVLDM	Mv$SL_{0/1}$-SVM	MvSTSVM	MDCTM
Ionosphere	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
Balance	98.91 ± 1.78	84.89 ± 9.46	93.66 ± 4.59	**99.62 ± 3.25**	95.50 ± 2.09	99.15 ± 1.28	95.12 ± 3.10	99.09 ± 1.35
100Leaves_1	**100.00 ± 0.00**	94.97 ± 3.70	95.34 ± 3.70	97.95 ± 1.87	98.38 ± 1.06	99.42 ± 2.53	98.82 ± 1.87	98.00 ± 2.09
100Leaves_2	98.98 ± 0.09	94.28 ± 2.20	96.65 ± 1.58	98.55 ± 3.02	98.99 ± 0.07	97.55 ± 3.72	98.50 ± 1.28	**99.00 ± 2.09**
100Leaves_3	**97.14 ± 3.93**	96.05 ± 2.01	96.09 ± 1.54	96.88 ± 3.95	96.05 ± 2.01	96.92 ± 1.18	96.20 ± 6.92	97.00 ± 1.12
Avg. Acc.	**98.76**	92.55	95.44	98.25	97.23	98.26	97.16	98.27
Avg. Rank	**2.25**	7.88	6.75	3.75	4.63	3.25	4.75	2.75

Table B.11
F1_score rates (%) with standard deviations for the UCI datasets.

Dataset	PSVM-2V	MV-LSSVM	MVLSSVM-2C	MvSLMC	MVLDM	Mv$SL_{0/1}$-SVM	MvSTSVM	MDCTM
Ionosphere	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
Balance	98.74 ± 1.33	82.93 ± 1.17	92.64 ± 5.67	99.23 ± 3.18	97.44 ± 1.29	98.83 ± 1.26	94.31 ± 2.98	**99.12 ± 1.28**
100Leaves_1	97.56 ± 2.87	94.64 ± 4.05	94.92 ± 4.31	97.37 ± 1.84	98.41 ± 1.00	**99.01 ± 2.50**	98.35 ± 1.91	98.49 ± 1.65
100Leaves_2	92.98 ± 4.86	93.99 ± 2.35	96.53 ± 1.53	97.94 ± 2.98	**98.51 ± 0.08**	96.82 ± 3.68	97.72 ± 1.30	98.24 ± 1.44
100Leaves_3	94.45 ± 4.34	96.05 ± 2.16	95.89 ± 1.93	95.89 ± 3.91	95.88 ± 2.16	96.40 ± 1.16	94.12 ± 6.86	**98.47 ± 0.58**
Avg. Acc.	95.93	91.86	94.99	97.61	97.56	97.76	96.12	**98.58**
Avg. Rank	6.00	7.38	5.88	3.37	3.63	2.75	5.25	**1.75**

Table B.12
Precision rates (%) with standard deviations for the Handwritten datasets.

Dataset	PSVM-2V	MV-LSSVM	MVLSSVM-2C	MvSLMC	MVLDM	Mv$SL_{0/1}$-SVM	MvSTSVM	MDCTM
Handwritten_1	97.18 ± 6.25	97.15 ± 4.64	97.08 ± 4.83	98.82 ± 3.18	92.86 ± 1.36	99.12 ± 1.69	**100.00 ± 0.00**	**100.00 ± 0.00**
Handwritten_2	99.00 ± 3.16	99.00 ± 3.10	97.92 ± 4.50	96.35 ± 1.67	**100.00 ± 0.00**	99.86 ± 4.10	98.02 ± 3.07	**100.00 ± 0.00**
Handwritten_3	97.50 ± 7.91	97.15 ± 4.64	94.42 ± 6.89	97.28 ± 2.77	99.05 ± 2.13	97.62 ± 2.42	97.98 ± 1.27	**100.00 ± 0.00**
Handwritten_4	**100.00 ± 0.00**	95.27 ± 6.34	97.15 ± 4.64	**100.00 ± 0.00**	95.96 ± 4.82	98.25 ± 1.83	96.85 ± 4.10	**100.00 ± 0.00**
Handwritten_5	**100.00 ± 0.00**	93.19 ± 10.03	96.39 ± 6.15	97.42 ± 3.59	94.68 ± 3.57	97.43 ± 2.62	98.43 ± 2.11	**100.00 ± 0.00**
Handwritten_6	98.45 ± 3.27	**100.00 ± 0.00**	**100.00 ± 0.00**	99.83 ± 2.52	**100.00 ± 0.00**	**100.00 ± 0.00**	99.82 ± 1.67	**100.00 ± 0.00**
Handwritten_7	**100.00 ± 0.00**	96.49 ± 5.84	**100.00 ± 0.00**	99.90 ± 1.80	99.00 ± 3.16	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
Handwritten_8	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
Handwritten_9	97.92 ± 4.50	93.05 ± 9.03	94.42 ± 8.93	95.82 ± 2.42	96.49 ± 5.84	98.18 ± 1.15	95.72 ± 4.21	**100.00 ± 0.00**
Avg. Acc.	99.11	96.77	97.54	98.33	98.15	98.92	98.35	**100.00**
Avg. Rank	3.94	6.38	5.44	5.31	4.50	3.44	4.81	**2.19**

Table B.13
Recall rates (%) with standard deviations for the Handwritten datasets.

Dataset	PSVM-2V	MV-LSSVM	MVLSSVM-2C	MvSLMC	MVLDM	Mv$SL_{0/1}$-SVM	MvSTSVM	MDCTM
Handwritten_1	**100.00 ± 0.00**	96.98 ± 7.40	98.46 ± 4.87	99.55 ± 3.27	90.00 ± 3.45	99.58 ± 1.73	**100.00 ± 0.00**	98.98 ± 3.22
Handwritten_2	98.18 ± 3.83	97.46 ± 5.50	99.23 ± 2.43	97.38 ± 1.71	**100.00 ± 0.00**	99.98 ± 4.21	98.93 ± 3.15	**100.00 ± 0.00**
Handwritten_3	**98.91 ± 0.23**	96.26 ± 8.14	98.46 ± 4.87	98.51 ± 2.85	98.00 ± 4.47	98.35 ± 2.48	98.62 ± 1.31	96.96 ± 2.11
Handwritten_4	**100.00 ± 0.00**	96.79 ± 6.76	96.98 ± 7.40	**100.00 ± 0.00**	99.09 ± 1.92	98.79 ± 1.88	98.10 ± 4.21	**100.00 ± 0.00**
Handwritten_5	98.42 ± 3.37	97.32 ± 4.35	96.98 ± 7.40	98.55 ± 3.68	99.57 ± 1.37	98.34 ± 2.68	99.05 ± 2.17	**100.00 ± 0.00**
Handwritten_6	**100.00 ± 0.00**	98.23 ± 3.77	98.18 ± 5.75	99.98 ± 2.58	**100.00 ± 0.00**	**100.00 ± 0.00**	99.95 ± 1.72	**100.00 ± 0.00**
Handwritten_7	**100.00 ± 0.00**	**100.00 ± 0.00**	99.09 ± 2.87	99.99 ± 1.84	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
Handwritten_8	**100.00 ± 0.00**	99.17 ± 2.64	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
Handwritten_9	99.23 ± 2.43	**100.00 ± 0.00**	99.23 ± 2.43	97.08 ± 2.48	99.23 ± 2.43	98.66 ± 1.18	97.18 ± 4.32	97.98 ± 1.34
Avg. Acc.	99.34	98.15	98.52	98.94	**99.49**	99.27	98.97	99.24
Avg. Rank	3.31	6.19	5.75	5.00	**3.25**	4.19	4.44	3.88

Table B.14
F1_score rates (%) with standard deviations for the Handwritten datasets.

Dataset	PSVM-2V	MV-LSSVM	MVLSSVM-2C	MvSLMC	MVLDM	Mv$SL_{0/1}$-SVM	MvSTSVM	MDCTM
Handwritten_1	98.47 ± 3.41	96.94 ± 5.28	97.70 ± 4.14	99.18 ± 3.22	91.35 ± 2.17	99.35 ± 1.71	**100.00 ± 0.00**	99.49 ± 1.87
Handwritten_2	98.52 ± 2.38	98.11 ± 3.15	97.54 ± 3.28	96.86 ± 1.69	**100.00 ± 0.00**	99.92 ± 4.15	98.47 ± 3.11	**100.00 ± 0.00**
Handwritten_3	**98.57 ± 4.52**	96.54 ± 5.49	96.24 ± 4.73	97.89 ± 2.81	98.46 ± 2.67	97.98 ± 2.45	98.30 ± 1.29	98.46 ± 2.67
Handwritten_4	**100.00 ± 0.00**	95.86 ± 5.23	96.94 ± 5.28	**100.00 ± 0.00**	97.42 ± 2.51	98.52 ± 1.85	97.47 ± 4.15	**100.00 ± 0.00**
Handwritten_5	99.18 ± 1.76	94.89 ± 5.83	96.61 ± 6.25	97.98 ± 3.63	97.02 ± 1.93	97.88 ± 2.65	98.74 ± 2.14	**100.00 ± 0.00**
Handwritten_6	99.19 ± 1.70	99.07 ± 1.98	99.00 ± 3.16	99.90 ± 2.55	**100.00 ± 0.00**	**100.00 ± 0.00**	99.88 ± 1.69	99.49 ± 0.98
Handwritten_7	**100.00 ± 0.00**	98.13 ± 3.12	99.52 ± 1.51	99.95 ± 1.82	99.47 ± 1.66	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
Handwritten_8	**100.00 ± 0.00**	99.57 ± 1.37	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
Handwritten_9	98.50 ± 2.51	96.18 ± 5.16	96.55 ± 5.30	96.44 ± 2.45	97.73 ± 3.11	98.42 ± 1.16	96.44 ± 4.26	**98.97 ± 1.19**
Avg. Acc.	99.24	97.29	97.80	98.62	98.76	99.09	98.66	**99.62**
Avg. Rank	2.94	7.50	6.50	4.81	4.06	3.50	4.25	**2.43**

Table B.15
Precision rates (%) with standard deviations for the AWA datasets.

Dataset	PSVM-2V	MV-LSSVM	MVLSSVM-2C	MvSLMC	MVLDM	Mv$SL_{0/1}$-SVM	MvSTSVM	MDCTM
AWA_1	88.05 ± 7.19	84.88 ± 9.60	84.90 ± 9.31	92.13 ± 6.32	86.63 ± 2.42	92.15 ± 4.21	92.45 ± 3.58	**95.93 ± 4.21**
AWA_2	78.91 ± 6.94	**100.00 ± 0.00**	**100.00 ± 0.00**	88.88 ± 3.67	99.43 ± 1.28	95.12 ± 2.98	94.87 ± 2.42	99.94 ± 0.54
AWA_3	89.22 ± 7.23	68.77 ± 5.47	73.74 ± 9.76	91.2 ± 4.13	**100.00 ± 0.00**	93.05 ± 3.82	86.32 ± 3.56	94.09 ± 1.68
AWA_4	83.22 ± 11.23	55.72 ± 12.85	63.32 ± 5.92	91.75 ± 2.97	84.00 ± 6.12	86.21 ± 1.45	81.99 ± 1.77	**100.00 ± 0.00**
AWA_5	77.92 ± 7.70	99.35 ± 1.27	99.35 ± 1.27	93.09 ± 3.55	98.43 ± 1.28	91.15 ± 1.72	88.15 ± 4.23	**99.98 ± 0.19**
AWA_6	90.75 ± 10.67	80.61 ± 3.28	75.32 ± 10.32	95.48 ± 3.15	**99.50 ± 1.12**	95.01 ± 7.32	87.62 ± 2.45	95.93 ± 3.56
AWA_7	85.61 ± 5.67	98.99 ± 1.45	85.53 ± 5.87	83.88 ± 2.41	92.29 ± 3.66	95.02 ± 1.11	95.71 ± 1.98	**99.49 ± 0.18**
AWA_8	81.72 ± 7.65	82.89 ± 4.94	85.53 ± 5.87	90.25 ± 7.32	84.06 ± 4.53	94.15 ± 3.72	90.83 ± 1.34	**94.92 ± 2.45**
AWA_9	84.35 ± 6.91	**99.23 ± 2.45**	82.22 ± 5.47	88.93 ± 1.98	98.90 ± 1.50	92.32 ± 2.78	93.92 ± 3.23	**99.23 ± 2.45**
AWA_10	85.46 ± 6.98	**99.35 ± 0.23**	85.02 ± 7.36	86.80 ± 5.26	83.98 ± 4.55	92.18 ± 2.65	88.91 ± 1.09	98.98 ± 0.26
AWA_11	92.25 ± 3.82	85.53 ± 5.87	84.30 ± 1.91	83.73 ± 5.34	78.31 ± 5.58	98.01 ± 1.05	94.12 ± 1.98	**99.00 ± 0.09**
AWA_12	83.91 ± 6.54	**99.13 ± 2.23**	84.06 ± 4.53	93.14 ± 5.67	85.53 ± 5.87	89.21 ± 4.78	90.88 ± 1.90	96.72 ± 3.44
AWA_13	80.64 ± 5.44	**98.34 ± 1.33**	**98.34 ± 1.33**	82.95 ± 1.52	87.20 ± 5.08	86.72 ± 6.21	86.54 ± 5.91	96.96 ± 2.33
AWA_14	86.47 ± 6.85	99.35 ± 1.98	99.35 ± 1.98	86.81 ± 7.54	96.81 ± 2.98	97.45 ± 0.18	93.89 ± 1.33	**100.00 ± 0.00**
AWA_15	85.58 ± 5.14	**100.00 ± 0.00**	99.35 ± 1.44	88.94 ± 4.23	96.08 ± 3.56	94.21 ± 2.01	97.91 ± 8.72	**100.00 ± 0.00**
AWA_16	90.52 ± 8.61	**98.32 ± 4.23**	94.61 ± 4.86	85.61 ± 8.92	89.71 ± 3.66	93.88 ± 3.12	92.67 ± 1.11	96.72 ± 3.45
AWA_17	84.58 ± 10.47	**100.00 ± 0.00**	95.24 ± 4.68	91.61 ± 7.89	96.61 ± 3.91	90.32 ± 2.52	86.11 ± 5.12	96.95 ± 4.23
AWA_18	89.55 ± 4.14	96.08 ± 5.19	96.17 ± 5.10	91.92 ± 9.28	88.17 ± 6.89	90.35 ± 3.82	91.75 ± 2.54	**100.00 ± 0.00**
AWA_19	96.90 ± 4.90	95.89 ± 3.62	97.10 ± 6.48	**97.79 ± 1.24**	93.51 ± 5.15	97.02 ± 1.28	94.52 ± 1.32	97.10 ± 2.53
AWA_20	91.41 ± 5.15	**97.54 ± 4.06**	95.75 ± 5.71	91.09 ± 1.31	97.26 ± 3.76	91.32 ± 4.91	92.99 ± 2.99	96.14 ± 2.16
AWA_21	92.82 ± 5.82	91.74 ± 5.73	89.12 ± 5.25	**98.07 ± 1.99**	95.53 ± 4.97	97.01 ± 0.85	94.67 ± 2.34	96.90 ± 4.90
AWA_22	92.72 ± 4.34	95.65 ± 3.22	**100.00 ± 0.00**	99.48 ± 1.15	96.80 ± 5.63	95.82 ± 1.91	96.88 ± 3.98	**100.00 ± 0.00**
AWA_23	96.81 ± 5.71	95.67 ± 5.81	95.67 ± 5.81	94.70 ± 1.90	95.51 ± 4.97	97.12 ± 2.05	94.15 ± 1.65	**100.00 ± 0.00**
AWA_24	96.08 ± 5.19	98.95 ± 3.33	95.51 ± 4.97	91.10 ± 5.90	97.10 ± 3.92	93.88 ± 1.45	93.25 ± 1.88	**100.00 ± 0.00**
AWA_25	93.25 ± 2.88	85.01 ± 9.52	87.21 ± 12.12	87.84 ± 5.20	93.51 ± 7.82	91.85 ± 2.54	91.05 ± 1.34	**100.00 ± 0.00**
AWA_26	85.90 ± 5.82	**100.00 ± 0.00**	**100.00 ± 0.00**	83.85 ± 4.92	**100.00 ± 0.00**	96.65 ± 0.85	93.15 ± 2.31	**100.00 ± 0.00**
AWA_27	**100.00 ± 0.00**	89.09 ± 9.21	85.01 ± 9.52	93.34 ± 1.82	88.67 ± 10.21	100.00 ± 0.00	96.75 ± 0.44	94.04 ± 5.23
AWA_28	82.28 ± 9.47	75.79 ± 8.08	84.01 ± 9.37	87.19 ± 7.01	94.88 ± 4.01	88.45 ± 2.86	91.25 ± 2.45	**94.96 ± 4.90**
AWA_29	93.52 ± 6.94	89.72 ± 7.81	79.61 ± 10.25	89.89 ± 9.99	87.59 ± 5.97	93.95 ± 1.56	94.25 ± 1.72	**99.47 ± 2.98**
AWA_30	99.09 ± 1.87	85.70 ± 9.21	81.25 ± 10.77	89.94 ± 4.86	91.70 ± 7.66	94.05 ± 2.12	95.88 ± 0.78	**99.49 ± 1.15**
AWA_31	92.47 ± 6.32	80.59 ± 7.19	91.05 ± 7.66	86.49 ± 7.11	86.25 ± 8.52	93.25 ± 2.75	**95.12 ± 1.47**	91.70 ± 7.66
AWA_32	96.75 ± 3.82	91.18 ± 7.66	89.27 ± 7.81	93.27 ± 6.85	88.82 ± 6.67	96.95 ± 0.95	94.05 ± 2.04	**100.00 ± 0.00**
AWA_33	**100.00 ± 0.00**	83.82 ± 3.86	87.49 ± 4.04	85.07 ± 6.21	99.49 ± 1.15	87.35 ± 3.01	89.35 ± 2.52	94.69 ± 4.26
AWA_34	81.70 ± 8.54	90.57 ± 7.04	**92.47 ± 6.32**	91.60 ± 3.05	84.25 ± 6.61	91.65 ± 1.70	91.15 ± 1.27	84.32 ± 7.23
AWA_35	89.27 ± 7.81	90.94 ± 7.23	**98.01 ± 2.18**	85.72 ± 3.71	97.49 ± 1.15	88.45 ± 2.35	89.55 ± 2.82	**98.01 ± 1.11**
AWA_36	**100.00 ± 0.00**	79.90 ± 18.70	79.90 ± 18.70	84.43 ± 3.81	91.18 ± 7.66	88.45 ± 2.35	89.55 ± 2.82	84.46 ± 5.32
AWA_37	92.59 ± 6.46	88.07 ± 7.44	**95.68 ± 5.34**	87.89 ± 8.04	92.71 ± 5.60	89.05 ± 1.39	90.15 ± 1.84	84.32 ± 3.56
AWA_38	85.26 ± 9.00	81.70 ± 2.64	85.54 ± 9.46	**98.44 ± 2.22**	92.26 ± 6.16	91.45 ± 2.06	91.55 ± 1.56	84.32 ± 5.32
AWA_39	89.49 ± 11.63	89.10 ± 6.55	79.12 ± 4.12	87.59 ± 7.84	89.62 ± 6.17	88.35 ± 2.68	**92.95 ± 0.81**	84.32 ± 3.44
AWA_40	88.79 ± 6.40	80.87 ± 7.89	72.55 ± 8.41	84.31 ± 3.98	92.40 ± 7.40	92.15 ± 1.60	**92.95 ± 2.16**	92.87 ± 3.67
AWA_41	**100.00 ± 0.00**	81.72 ± 7.56	84.01 ± 4.72	91.19 ± 7.19	**100.00 ± 0.00**	95.88 ± 0.66	91.25 ± 2.92	**100.00 ± 0.00**
AWA_42	81.69 ± 2.39	79.88 ± 5.31	74.97 ± 7.62	84.65 ± 7.49	83.82 ± 3.86	**86.35 ± 2.62**	85.45 ± 3.04	84.32 ± 9.23
AWA_43	88.47 ± 7.70	73.45 ± 8.01	83.45 ± 6.06	90.11 ± 12.97	89.27 ± 7.81	**95.45 ± 1.11**	93.15 ± 1.91	84.32 ± 10.24
AWA_44	82.84 ± 5.90	89.81 ± 7.61	83.61 ± 5.70	92.61 ± 2.47	**100.00 ± 0.00**	91.25 ± 2.31	89.25 ± 1.65	**100.00 ± 0.00**
AWA_45	78.32 ± 8.68	82.57 ± 6.39	80.46 ± 7.29	86.11 ± 9.21	84.60 ± 7.13	**91.45 ± 1.29**	90.15 ± 2.38	84.32 ± 11.24
Avg. Acc.	89.23	89.08	87.93	87.74	92.34	92.83	91.80	**95.41**
Avg. Rank	5.40	4.98	5.27	5.16	4.32	3.89	4.34	**2.64**

Table B.16
Recall rates (%) with standard deviations for the AWA datasets.

Dataset	PSVM-2V	MV-LSSVM	MVLSSVM-2C	MvSLMC	MVLDM	Mv$SL_{0/1}$-SVM	MvSTSVM	MDCTM
AWA_1	81.56 ± 7.97	86.92 ± 4.95	98.00 ± 2.09	92.00±1.76	**100.00 ± 0.00**	89.87 ± 5.67	90.12±4.12	94.97 ± 0.18
AWA_2	87.48 ± 10.24	78.89 ± 0.62	78.89 ± 0.62	**99.00±1.36**	7.00 ± 2.74	93.21 ± 3.87	93.21±1.93	98.49 ± 0.53
AWA_3	84.72 ± 6.99	64.26 ± 5.59	82.57 ± 8.40	92.00±3.79	71.00 ± 15.47	90.45 ± 4.91	85.44±2.22	**93.92 ± 1.54**
AWA_4	85.19 ± 4.73	89.56 ± 23.56	82.05 ± 6.64	96.00±1.06	**99.50 ± 1.12**	83.12 ± 2.32	80.12±2.89	76.88 ± 3.32
AWA_5	94.70 ± 7.04	77.88 ± 10.79	77.88 ± 10.79	**99.00±2.23**	87.50 ± 2.50	88.67 ± 2.45	87.33±3.22	97.98 ± 3.45
AWA_6	86.59 ± 8.75	**99.00 ± 2.24**	**99.00 ± 1.37**	94.23±5.18	96.00 ± 3.35	92.34 ± 8.67	86.01±5.97	94.97 ± 1.34
AWA_7	**99.57 ± 1.37**	78.89 ± 7.22	98.55 ± 2.35	97.20±3.24	98.00 ± 1.37	92.88 ± 1.54	94.88±2.13	98.68 ± 0.37
AWA_8	83.62 ± 11.00	**99.67 ± 0.10**	98.50 ± 2.24	97.11±1.37	98.50 ± 1.37	91.32 ± 4.65	89.67±3.91	93.96 ± 3.45
AWA_9	79.33 ± 8.38	76.88 ± 7.56	94.26 ± 6.78	**100.00±0.00**	90.00 ± 3.95	89.78 ± 3.45	92.45±3.32	76.88 ± 7.56
AWA_10	98.19 ± 3.18	77.88 ± 4.93	98.55 ± 2.35	**100.00±0.00**	98.00 ± 2.09	89.65 ± 3.32	87.22±1.22	97.98 ± 3.44
AWA_11	96.87 ± 4.80	98.50 ± 2.24	99.00 ± 1.37	96.00±3.71	98.50 ± 1.12	96.32 ± 1.42	93.55±1.11	**99.49 ± 0.12**
AWA_12	84.38 ± 6.54	76.88 ± 7.56	**98.50 ± 1.37**	95.00±10.95	**98.50 ± 2.24**	86.45 ± 5.87	89.34±5.31	88.94 ± 4.32
AWA_13	90.42 ± 7.36	77.89 ± 3.23	77.89 ± 3.23	97.00±1.74	**100.00 ± 0.00**	83.21 ± 7.45	85.67±1.23	96.48 ± 2.12
AWA_14	91.29 ± 4.38	77.88 ± 12.34	77.88 ± 12.34	99.00±3.35	90.50 ± 6.47	96.12 ± 0.25	92.78±7.31	**100.00 ± 0.00**
AWA_15	**100.00 ± 0.00**	79.39 ± 4.56	80.50 ± 6.94	99.00±1.37	92.50 ± 5.59	91.78 ± 2.54	97.32±2.12	**100.00 ± 0.00**
AWA_16	**91.52 ± 8.02**	76.88 ± 4.21	83.25 ± 4.38	97.00±5.18	91.50 ± 4.18	91.32 ± 3.87	91.23±2.31	88.94 ± 6.45
AWA_17	86.92 ± 12.11	77.88 ± 9.43	72.39 ± 6.47	**98.00±3.26**	81.77 ± 3.67	87.78 ± 3.21	84.89±4.98	95.97 ± 4.22
AWA_18	93.07 ± 5.33	87.62 ± 5.26	89.62 ± 6.61	99.00±1.37	92.33 ± 5.93	87.32 ± 4.65	90.12±1.96	**100.00 ± 0.00**
AWA_19	95.40 ± 3.65	82.29 ± 3.59	83.83 ± 6.48	89.00±4.18	93.00 ± 4.81	95.32 ± 1.65	93.88±1.34	**98.50 ± 1.37**
AWA_20	92.51 ± 7.64	79.19 ± 5.91	88.99 ± 5.73	92.00±2.74	90.57 ± 6.63	88.45 ± 5.98	91.45±3.13	**99.00 ± 2.24**
AWA_21	86.70 ± 7.68	85.12 ± 4.62	89.80 ± 7.02	**100.00±0.00**	89.36 ± 5.90	95.21 ± 1.12	93.12±4.98	95.40 ± 3.65
AWA_22	81.46 ± 7.44	**99.49 ± 1.10**	78.89 ± 5.32	92.00±2.24	92.38 ± 5.37	93.67 ± 2.45	96.01±2.21	**99.49 ± 0.23**
AWA_23	93.47 ± 3.57	87.62 ± 5.26	87.62 ± 5.26	**98.00±3.06**	89.50 ± 6.43	95.12 ± 2.58	91.78 ± 2.10	97.00 ± 1.12
AWA_24	87.62 ± 5.26	79.07 ± 6.47	89.50 ± 6.43	**99.00±1.37**	83.83 ± 6.48	91.78 ± 2.10	90.65 ± 2.41	89.50 ± 6.22
AWA_25	89.31 ± 4.46	88.50 ± 6.53	**98.00 ± 2.74**	96.00±3.79	97.00 ± 2.09	89.25 ± 3.25	88.45 ± 1.71	96.50 ± 2.85
AWA_26	98.50 ± 2.24	76.00 ± 17.55	76.00 ± 17.55	100.00±0.00	80.00 ± 13.58	94.88 ± 1.09	90.45 ± 2.96	**100.00 ± 0.00**
AWA_27	87.00 ± 5.42	87.67 ± 6.00	88.50 ± 6.53	98.00±1.76	99.00 ± 1.37	**99.50 ± 0.50**	95.12 ± 0.56	97.50 ± 1.77
AWA_28	88.54 ± 7.05	80.09 ± 8.16	91.76 ± 4.83	87.00±6.02	98.00 ± 2.09	85.32 ± 3.65	88.45 ± 3.12	**98.50 ± 1.37**
AWA_29	**99.09 ± 1.92**	90.31 ± 6.16	94.65 ± 4.32	99.00±2.23	99.00 ± 1.37	91.78 ± 1.99	91.88 ± 2.19	94.97 ± 5.11
AWA_30	90.00 ± 11.86	92.38 ± 5.84	**99.57 ± 1.37**	87.00±5.57	95.31 ± 4.59	91.45 ± 2.71	94.12 ± 1.00	92.50 ± 7.29
AWA_31	87.28 ± 12.07	98.56 ± 2.32	86.38 ± 10.45	91.50±3.54	**100.00 ± 0.00**	90.45 ± 3.51	93.12 ± 1.88	95.31 ± 4.59
AWA_32	99.09 ± 1.92	84.95 ± 10.95	90.31 ± 6.16	99.00±1.37	99.09 ± 1.92	95.12 ± 1.21	91.45 ± 2.61	**100.00 ± 0.00**
AWA_33	91.50 ± 8.02	99.00 ± 1.45	99.02 ± 1.10	99.00±2.23	94.00 ± 2.85	91.45 ± 3.09	89.25 ± 3.54	**99.57 ± 1.37**
AWA_34	87.06 ± 7.06	86.86 ± 10.76	87.28 ± 12.07	**100.00±0.00**	99.50 ± 1.12	84.12 ± 3.85	86.12 ± 3.21	**100.00 ± 0.00**
AWA_35	90.31 ± 6.16	86.88 ± 11.20	83.50 ± 9.62	96.00±4.41	94.00 ± 2.85	88.95 ± 2.17	88.55 ± 1.62	**98.00 ± 1.12**
AWA_36	78.78 ± 9.23	94.99 ± 6.41	94.99 ± 6.41	95.00±4.81	84.95 ± 10.59	85.32 ± 3.00	86.32 ± 3.61	**100.00 ± 0.00**
AWA_37	99.57 ± 1.37	90.84 ± 5.80	83.80 ± 6.96	95.00±5.30	97.27 ± 4.02	86.25 ± 1.78	87.35 ± 2.35	**100.00 ± 0.00**
AWA_38	88.05 ± 6.14	**100.00 ± 0.00**	92.22 ± 4.47	83.00±12.74	86.30 ± 12.30	88.75 ± 2.63	88.85 ± 1.99	**100.00 ± 0.00**
AWA_39	77.53 ± 9.48	84.30 ± 6.13	90.97 ± 8.54	84.00±14.42	87.04 ± 7.67	85.12 ± 3.43	91.12 ± 1.03	**100.00 ± 0.00**
AWA_40	98.50 ± 1.37	83.62 ± 11.00	97.71 ± 5.58	**99.00±1.37**	72.00 ± 13.16	89.55 ± 2.05	90.45 ± 2.75	98.30 ± 2.84
AWA_41	80.00 ± 13.58	83.62 ± 11.00	91.74 ± 8.86	83.00±10.81	86.42 ± 2.34	**94.12 ± 0.84**	88.45 ± 3.73	78.39 ± 13.53
AWA_42	**100.00 ± 0.00**	94.30 ± 6.87	96.32 ± 5.40	89.00±10.24	**100.00 ± 0.00**	83.12 ± 3.35	82.12 ± 3.88	**100.00 ± 0.00**
AWA_43	90.31 ± 6.16	97.27 ± 5.56	88.99 ± 0.66	96.00±10.06	90.31 ± 6.16	93.55 ± 1.42	90.45 ± 2.44	**100.00 ± 0.00**
AWA_44	**90.81 ± 5.87**	88.00 ± 6.60	88.99 ± 0.66	94.00±3.79	88.00 ± 6.47	88.45 ± 2.96	86.45 ± 2.11	78.39 ± 19.23
AWA_45	85.50 ± 10.98	87.41 ± 6.98	83.64 ± 7.29	97.00±5.42	98.00 ± 2.74	88.75 ± 1.65	87.35 ± 3.05	**100.00 ± 0.00**
Avg. Acc.	90.18	86.00	89.91	95.27	92.16	90.38	89.86	**95.60**
Avg. Rank	4.46	6.01	5.02	3.25	4.10	5.01	5.19	**2.76**

Table B.17
F1_score rates (%) with standard deviations for the AWA datasets.

Dataset	PSVM-2V	MV-LSSVM	MVLSSVM-2C	MvSLMC	MVLDM	Mv$SL_{0/1}$-SVM	MvSTSVM	MDCTM
AWA_1	84.32 ± 4.83	80.38 ± 5.52	64.78 ± 7.45	91.81 ± 1.69	92.82 ± 1.40	90.98 ± 4.89	91.27 ± 2.41	**95.45 ± 1.57**
AWA_2	82.64 ± 6.73	88.20 ± 4.62	88.20 ± 4.62	93.65 ± 2.39	92.78 ± 1.85	94.15 ± 3.42	94.03 ± 1.35	**98.98 ± 0.26**
AWA_3	86.73 ± 5.85	61.06 ± 6.91	77.56 ± 7.76	91.18 ± 2.86	82.19 ± 11.75	91.72 ± 4.36	85.88 ± 6.38	**93.98 ± 1.58**
AWA_4	83.69 ± 6.98	65.00 ± 12.05	71.30 ± 4.88	**92.32 ± 1.06**	90.98 ± 3.17	84.63 ± 1.88	81.05 ± 3.62	86.93 ± 4.52
AWA_5	85.12 ± 4.85	87.32 ± 2.83	87.32 ± 2.83	95.91 ± 1.92	95.45 ± 2.26	89.89 ± 2.08	87.74 ± 1.11	**98.84 ± 2.45**
AWA_6	88.09 ± 6.87	88.82 ± 1.89	85.19 ± 6.04	94.41 ± 3.13	**97.69 ± 1.73**	93.65 ± 7.98	86.81 ± 5.41	95.45 ± 1.56
AWA_7	91.96 ± 3.20	88.20 ± 6.23	91.35 ± 3.82	90.58 ± 1.84	90.64 ± 2.51	93.93 ± 1.32	95.29 ± 2.35	**98.99 ± 0.76**
AWA_8	82.06 ± 6.40	90.58 ± 2.99	91.50 ± 4.12	93.16 ± 4.42	94.21 ± 2.46	92.70 ± 4.18	90.25 ± 4.20	**94.44 ± 2.44**
AWA_9	81.52 ± 6.27	86.93 ± 2.46	87.58 ± 3.92	**94.13 ± 1.10**	90.38 ± 2.69	91.02 ± 3.11	93.15 ± 1.34	86.93 ± 2.46
AWA_10	91.16 ± 3.20	87.32 ± 2.83	91.08 ± 3.96	92.94 ± 2.87	83.23 ± 3.23	90.89 ± 2.98	88.06 ± 3.30	**98.48 ± 3.65**
AWA_11	94.57 ± 3.12	91.05 ± 4.12	91.04 ± 0.84	89.10 ± 2.84	87.57 ± 3.72	97.15 ± 1.23	93.83 ± 2.78	**99.24 ± 0.15**
AWA_12	83.95 ± 6.45	86.93 ± 2.32	90.64 ± 2.51	93.59 ± 7.43	91.50 ± 4.12	87.80 ± 5.32	90.10 ± 4.51	**92.67 ± 5.23**
AWA_13	84.93 ± 3.35	87.57 ± 3.23	87.57 ± 3.23	89.40 ± 1.03	93.10 ± 2.93	84.91 ± 6.82	86.1 ± 2.57	**96.72 ± 3.32**
AWA_14	88.59 ± 3.28	87.32 ± 3.19	87.32 ± 3.19	92.17 ± 4.81	93.46 ± 4.09	96.78 ± 0.21	93.33 ± 1.24	**100.00 ± 0.00**
AWA_15	92.16 ± 2.95	88.51 ± 5.34	88.82 ± 4.51	93.66 ± 2.56	94.10 ± 2.23	92.97 ± 2.27	97.61 ± 4.30	**100.00 ± 0.00**
AWA_16	90.55 ± 5.11	86.93 ± 5.30	88.46 ± 3.37	90.54 ± 5.51	90.59 ± 3.37	92.57 ± 3.48	91.94 ± 5.42	**92.67 ± 1.05**
AWA_17	84.72 ± 7.18	87.57 ± 5.34	82.07 ± 4.62	94.58 ± 5.10	88.49 ± 2.65	89.02 ± 2.86	85.5 ± 3.45	**96.46 ± 3.25**
AWA_18	91.15 ± 3.12	91.54 ± 3.11	92.63 ± 4.56	95.18 ± 5.68	89.94 ± 4.21	88.79 ± 4.21	90.93 ± 1.99	**100.00 ± 0.00**
AWA_19	96.06 ± 3.25	88.51 ± 2.93	89.82 ± 4.15	93.16 ± 2.63	93.24 ± 4.80	96.15 ± 1.46	94.2 ± 2.46	**97.78 ± 1.33**
AWA_20	91.78 ± 4.95	87.22 ± 3.35	92.09 ± 4.26	91.53 ± 1.72	93.67 ± 4.28	89.85 ± 5.42	92.21 ± 3.96	**97.54 ± 1.77**
AWA_21	86.70 ± 7.68	85.12 ± 4.62	89.80 ± 7.02	98.76 ± 0.87	89.36 ± 5.90	96.09 ± 0.98	93.89 ± 3.51	**96.06 ± 3.65**
AWA_22	81.46 ± 7.44	**99.49 ± 1.10**	78.89 ± 5.32	95.30 ± 0.78	94.72 ± 2.18	96.44 ± 5.15	92.38 ± 5.37	**99.74 ± 0.23**
AWA_23	93.47 ± 3.57	87.62 ± 5.26	87.62 ± 5.26	96.05 ± 1.66	89.50 ± 6.43	96.27 ± 2.30	92.95 ± 1.87	**98.47 ± 1.12**
AWA_24	87.62 ± 5.26	79.07 ± 6.47	89.50 ± 6.43	**94.82 ± 3.44**	83.83 ± 6.48	92.81 ± 1.64	91.93 ± 2.14	94.37 ± 6.22
AWA_25	89.31 ± 4.46	88.50 ± 6.53	98.00 ± 2.74	91.68 ± 3.94	97.00 ± 2.09	90.52 ± 2.88	89.72 ± 1.52	**98.20 ± 2.85**
AWA_26	98.50 ± 2.24	76.00 ± 17.55	76.00 ± 17.55	90.94 ± 2.76	80.00 ± 13.58	95.76 ± 0.96	91.76 ± 2.63	**100.00 ± 0.00**
AWA_27	87.00 ± 5.42	87.67 ± 6.00	88.50 ± 6.53	95.36 ± 1.00	99.00 ± 1.37	**99.75 ± 0.25**	95.93 ± 0.50	95.65 ± 1.77
AWA_28	88.54 ± 7.05	80.09 ± 8.16	91.76 ± 4.83	86.75 ± 5.80	98.00 ± 2.09	86.85 ± 3.25	89.82 ± 2.78	**96.62 ± 1.37**
AWA_29	**99.09 ± 1.92**	90.31 ± 6.16	94.65 ± 4.32	93.83 ± 6.27	99.00 ± 1.37	92.85 ± 1.77	93.05 ± 1.95	97.17 ± 5.11
AWA_30	90.00 ± 11.86	92.38 ± 5.84	**99.57 ± 1.37**	88.02 ± 3.38	95.31 ± 4.59	92.73 ± 2.40	94.99 ± 0.89	95.74 ± 7.29
AWA_31	89.30 ± 7.29	88.49 ± 4.31	88.22 ± 6.72	92.23 ± 4.44	92.41 ± 5.01	91.82 ± 3.12	**94.11 ± 1.67**	93.35 ± 5.45
AWA_32	97.86 ± 2.06	87.57 ± 7.26	89.52 ± 4.79	95.58 ± 4.47	93.55 ± 3.92	96.03 ± 1.08	92.73 ± 2.32	**100.00 ± 0.00**
AWA_33	95.41 ± 4.61	91.16 ± 2.27	93.29 ± 2.33	95.98 ± 4.45	96.64 ± 1.51	92.73 ± 2.75	90.68 ± 3.15	**97.01 ± 2.34**
AWA_34	84.08 ± 6.65	88.24 ± 6.69	89.30 ± 7.29	**91.59 ± 3.05**	91.20 ± 4.07	85.70 ± 3.42	87.68 ± 2.86	91.49 ± 5.98
AWA_35	89.52 ± 4.79	88.43 ± 7.21	90.28 ± 4.82	90.31 ± 3.37	96.64 ± 1.51	90.28 ± 1.93	89.82 ± 1.44	**98.00 ± 1.12**
AWA_36	88.13 ± 4.78	85.26 ± 10.24	85.26 ± 10.24	89.14 ± 3.74	88.27 ± 7.26	86.85 ± 2.67	87.89 ± 3.21	**91.49 ± 4.09**
AWA_37	**95.83 ± 3.58**	89.17 ± 4.33	89.16 ± 4.64	91.93 ± 6.27	94.87 ± 4.08	87.62 ± 1.58	88.76 ± 2.09	91.49 ± 5.32
AWA_38	86.41 ± 6.49	89.91 ± 1.60	88.42 ± 4.82	89.25 ± 7.03	88.70 ± 7.50	90.08 ± 2.34	90.20 ± 1.77	**91.49 ± 5.32**
AWA_39	82.27 ± 5.80	86.47 ± 4.93	84.41 ± 4.43	84.61 ± 7.09	88.11 ± 5.33	86.70 ± 3.05	**92.03 ± 0.92**	91.49 ± 9.74
AWA_40	93.26 ± 3.25	81.68 ± 6.83	82.95 ± 5.88	90.82 ± 2.69	93.91 ± 2.51	90.82 ± 1.82	91.67 ± 2.45	**95.49 ± 3.43**
AWA_41	88.37 ± 8.66	82.06 ± 6.04	87.34 ± 3.72	84.74 ± 8.46	**94.99 ± 0.75**	89.82 ± 3.32	83.78 ± 3.45	87.88 ± 7.23
AWA_42	89.91 ± 1.44	86.19 ± 2.90	84.02 ± 5.00	86.54 ± 7.26	91.16 ± 2.27	84.72 ± 2.98	**91.76 ± 2.17**	91.49 ± 5.24
AWA_43	89.09 ± 4.49	83.41 ± 5.66	85.82 ± 3.39	92.31 ± 10.06	89.52 ± 4.97	**94.49 ± 1.26**	91.76 ± 2.17	91.49 ± 7.23
AWA_44	90.03 ± 4.41	85.06 ± 3.64	85.91 ± 3.06	93.28 ± 2.88	**93.52 ± 3.69**	89.82 ± 2.63	87.82 ± 1.88	87.88 ± 8.22
AWA_45	81.65 ± 9.22	84.60 ± 3.90	81.73 ± 3.47	91.04 ± 6.34	90.67 ± 4.45	90.08 ± 1.47	88.73 ± 2.71	**91.49 ± 9.74**
Avg. Acc.	89.41	86.41	87.60	92.09	91.67	91.58	90.81	**95.22**
Avg. Rank	5.23	6.75	3.65	6.13	2.48	4.09	4.41	**1.72**

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